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---

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**32384 CHARLES COUNTY POMONKEY PUMP STATION  
IMPROVEMENTS**

**INSTRUMENTATION RECORD SUBMITTAL**

**SECTION 17500 - INSTRUMENTATION**

**June 05, 2026**

**ENGINEER: WRA**

**CONTRACTOR: JOHNSTON CONSTRUCTION**

***MANUFACTURERS' REPRESENTATIVE FOR ENGINEERED INSTRUMENTATION,  
PROCESS EQUIPMENT AND PUMPING SYSTEMS***

***Branch Offices: RICHMOND, VA and LEOLA, PA***

# **32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS**

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## **32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS**

### **General Comments:**

1. This submittal assumes cable lengths for the various instruments included.  
Contractor to confirm cable lengths for instruments.

### **Resubmittal #17500-01-01 Comment Confirmations:**

1. Specification 17500, Section 1.03 - B requires additional information beyond beam angle, width, and depth. This includes but is not limited to distance from "ladders, fill stream, floats, and any other nearby object".

Further, current dimensional indicate mounting of transmitter near fill stream. Per manufacturer instructions, transmitter shall not be installed above the filling curtain. Contractor shall provide dimensional drawings meeting the specification requirements, and manufacturer recommended installation requirements. Return record copy.

### **Sherwood Logan Response:**

Level sensor model has been changed to the 3" antenna to allow for a tighter beam angle. Updated dimensional drawings have been included. Additional specific mounting information will be coordinated with the contractor.

2. Submittal changed "LSL-02" tag to "LSL-03", and "LSH=-2" tag to "LSH-03". This still results in eight total floats submitted for this project. Per prior comment, only six are required.

Since Back Up Pump submittal is approved, and includes two floats, remove LSL-03 and LSH-03 from this submittal. Return record copy.

### **Sherwood Logan Response:**

Floats LSL-03 and LSH-03 have been removed from this submittal as

requested.

3. Mounting hardware shown on I-9 is not limited to external weight, but also cable strain reliefs, hook, cable eye, and anchor. Submit hardware as requested in record copy.

**Sherwood Logan Response:**

A float bracket has been submitted with included strain relief. Verify this mounting hardware is sufficient.

### Parts Index:

| SHT | MANUFACTURER      | MODEL                     | DESCRIPTION                          |
|-----|-------------------|---------------------------|--------------------------------------|
| 1   | Endress + Hauser  | FMR20B-FBBADTBNVCGXR0D+Z1 | Radar Level Transmitter              |
| 1   | Endress + Hauser  | 71429910                  | Swiveling Mounting Bracket,<br>Sewer |
| 2   | Anchor Scientific | GSE60NONC-GOLD            | Float Switch, NO                     |
| 2   | SJE Rhombus       | 1009433                   | Float Switch Mounting Bracket        |
| 4   | E+H               | RIA452-C211A11A           | Digital Process Meter Display        |

# 32384 CHARLES COUNTY POMONKEY PUMP STATION IMPROVEMENTS

## Data Sheet 1

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer:  | Johnston Construction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Reference: | 32384 Charles County Pomonkey Pump Station Improvements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Date:      | 06/05/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <u>Qty</u> | <u>Description</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 1          | <p>Mfg: Endress + Hauser: Radar Level Transmitter<br/>Model Number: FMR20B-FBBADTBNVCGXR0D+Z1</p> <p>Tags/Service:<br/>LT-01 / Pump Station Wet Well</p> <p>Specifications:</p> <ul style="list-style-type: none"> <li>• Up to 98.4ft measuring range</li> <li>• FMR20 - Micropilot FMR20 series time-of-flight radar measurement</li> <li>• FB - CAN/US IS Cl.I,II,III Div.1 Gr.A-G T4 AEx/Ex ia IIC T4, AEx/Ex ia IIIC</li> <li>• BA - 2 wire, 4-20mA HART output</li> <li>• D - LED + bluetooth display/operation</li> <li>• T - Pre-installed cable for electrical connection</li> <li>• BN - PVDF encapsulated, 80mm / 3" antenna</li> <li>• VCG - Thread ASME MNPT1 process connection cable entry</li> <li>• XR0 - Without process connection on antenna end, prepared for UNI slip on flange</li> <li>• D - 20m/65ft cable length</li> <li>• Z1 - Tagging</li> </ul> |
| 1          | <p>Mfg: Endress + Hauser: Swiveling Mounting Bracket, Sewer<br/>Model Number: 71429910</p> <p>Tags/Service:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|  |                                                                                                                                                                                                                                         |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>LT-01 / Pump Station Wet Well</p> <p>Specifications:</p> <ul style="list-style-type: none"><li>• Pivotal and adjustable arm for mounting E+H FMR20B</li><li>• Allows alignment of transmitter with the center of a channel</li></ul> |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## RADAR LEVEL TRANSMITTER

TAGS: LT-01

# Technical Information Micropilot FMR20B

Free-space radar  
HART



## Level measurement in liquids and bulk solids

### Application

- Continuous, non-contact level measurement of liquids and bulk solids
- Degree of protection: IP66/68 / NEMA Type 4X/6P
- Maximum measuring range up to 30 m (98 ft)
- Process temperature: -40 to 80 °C (-40 to 176 °F)
- Process pressure: -1 to 3 bar (-14 to 43 psi)
- Accuracy: Up to  $\pm 2$  mm (0.08 in)
- International explosion protection certificates

### Your benefits

- LED indicator for fast status detection
- Easy, guided commissioning with intuitive user interface
- Radar measuring device with *Bluetooth*® wireless technology and HART communication
- Easy, reliable and encrypted wireless remote access – ideal for difficult-to-reach installations, even in hazardous areas
- Commissioning, operation and maintenance via free iOS/Android app SmartBlue – saves time and reduces costs
- Flow measurement in open channels or weirs with totalizer

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# About this document

|         |                                                                                                                                                                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Symbols | <b>Safety symbols</b>                                                                                                                                                                                                                                                                   |
|         | <b> DANGER</b><br>This symbol alerts you to a dangerous situation. Failure to avoid this situation will result in serious or fatal injury.                                                             |
|         | <b> WARNING</b><br>This symbol alerts you to a potentially dangerous situation. Failure to avoid this situation can result in serious or fatal injury.                                                 |
|         | <b> CAUTION</b><br>This symbol alerts you to a potentially dangerous situation. Failure to avoid this situation can result in minor or medium injury.                                                  |
|         | <b>NOTICE</b><br>This symbol alerts you to a potentially harmful situation. Failure to avoid this situation can result in damage to the product or something in its vicinity.                                                                                                           |
|         | <b>Communication-specific symbols</b>                                                                                                                                                                                                                                                   |
|         | <b>Bluetooth®:</b> <br>Wireless data transmission between devices over a short distance via radio technology.                                                                                          |
|         | <b>Symbols for certain types of information</b>                                                                                                                                                                                                                                         |
|         | <i>Permitted:</i> <br>Procedures, processes or actions that are permitted.                                                                                                                             |
|         | <i>Forbidden:</i> <br>Procedures, processes or actions that are forbidden.                                                                                                                           |
|         | <i>Additional information:</i>                                                                                                                                                                       |
|         | <i>Reference to documentation:</i>                                                                                                                                                                   |
|         | <i>Reference to page:</i>                                                                                                                                                                            |
|         | <i>Series of steps:</i>  ,  ,  |
|         | <i>Result of an individual step:</i>                                                                                                                                                                 |
|         | <b>Symbols in graphics</b>                                                                                                                                                                                                                                                              |
|         | <i>Item numbers:</i> 1, 2, 3 ...                                                                                                                                                                                                                                                        |
|         | <i>Series of steps:</i>  ,  ,  |
|         | <i>Views:</i> A, B, C, ...                                                                                                                                                                                                                                                              |
|         | <b>List of abbreviations</b>                                                                                                                                                                                                                                                            |
|         | <b>PN</b><br>Nominal pressure                                                                                                                                                                                                                                                           |
|         | <b>MWP</b><br>Maximum working pressure<br>The maximum working pressure is indicated on the nameplate.                                                                                                                                                                                   |
|         | <b>ToF</b><br>Time of Flight                                                                                                                                                                                                                                                            |
|         | <b>DTM</b><br>Device Type Manager                                                                                                                                                                                                                                                       |
|         | <b>ε<sub>r</sub> (Dk value)</b><br>Relative dielectric constant                                                                                                                                                                                                                         |
|         | <b>Operating tool</b>                                                                                                                                                                                                                                                                   |

The term "operating tool" is used in place of the following operating software:

- FieldCare / DeviceCare, for operation via HART communication, IO-Link communication and PC
- SmartBlue app for operation using an Android or iOS smartphone or tablet

#### PLC

Programmable logic controller (PLC)

#### Graphic conventions

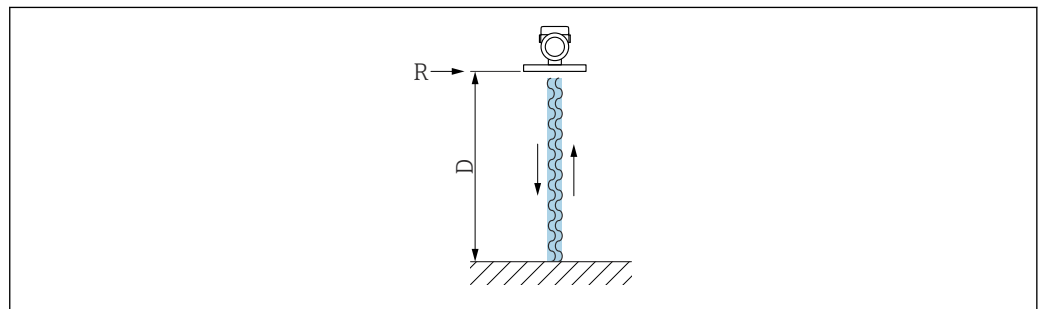


- Installation, explosion and electrical connection drawings are presented in simplified format
- Devices, assemblies, components and dimensional drawings are presented in reduced-line format
- Dimensional drawings are not to-scale representations; the dimensions indicated are rounded off to 2 decimal places
- Unless otherwise described, flanges are presented with sealing surface form EN 1092-1; ASME B16.5, RF.

## Function and system design

#### Measuring principle

The Micropilot is a "downward-looking" measuring system, operating based on the frequency modulated continuous wave method (FMCW). The antenna emits an electromagnetic wave at a continuously varying frequency. This wave is reflected by the product and received again by the antenna.



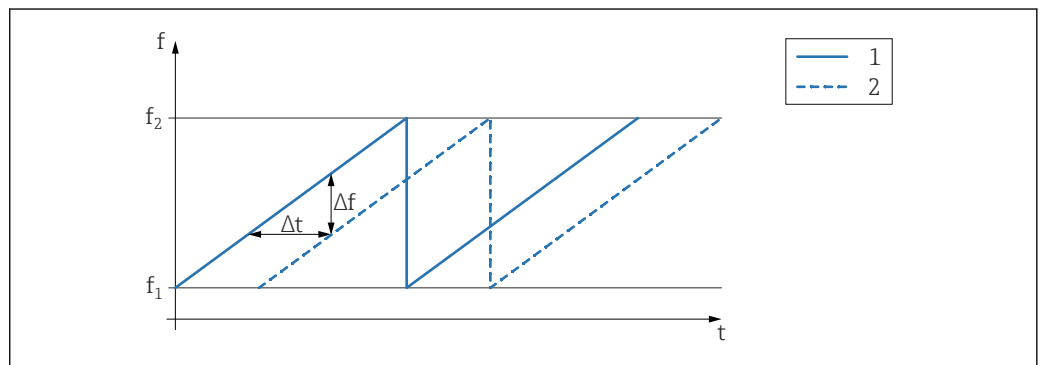
A0032017

1 FMCW principle: Transmission and reflection of the continuous wave

R Reference point of measurement

D Distance between reference point and product surface

The frequency of this wave is modulated in the form of a sawtooth signal between two limit frequencies  $f_1$  and  $f_2$ :



A0023771

2 FMCW principle: Result of frequency modulation

1 Transmitted signal

2 Received signal

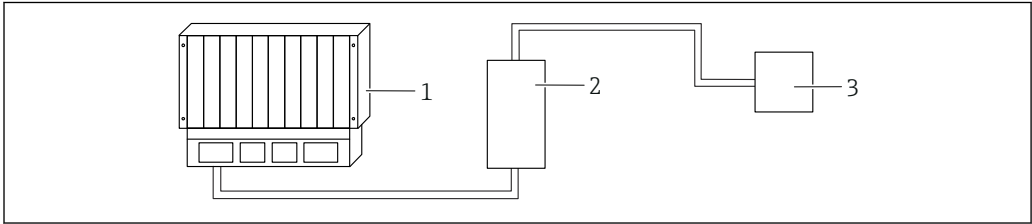
This results in the following difference frequency at any time between the transmitted signal and the received signal:

$\Delta f = k \Delta t$

where  $\Delta t$  is the run time and  $k$  is the specified increase in frequency modulation.  
 $\Delta t$  is given by the distance  $D$  between the reference point  $R$  and the product surface:

$D = (c \Delta t) / 2$

where  $c$  is the wave velocity.  
In summary,  $D$  can be calculated from the measured difference frequency  $\Delta f$ .  $D$  is then used to determine the fill level or flow rate.

|                  |                                                                                                                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Measuring system | <p>A complete measuring system comprises:</p> <div><p>A0053220</p></div> <div><p>1    PLC (programmable logic controller)</p><p>2    RMA42/RIA45 (if necessary)</p><p>3    Device</p></div> |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                   |                                                                                                                                                                                |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Communication and data processing | <ul style="list-style-type: none"><li>■ 4 to 20 mA with superimposed digital communication protocol HART, 2-wire</li><li>■ Bluetooth® wireless technology (optional)</li></ul> |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependability | <p><b>IT security</b></p> <p>The manufacturer warranty is valid only if the product is installed and used as described in the Operating Instructions. The product is equipped with security mechanisms to protect it against any inadvertent changes to the settings.</p> <p>IT security measures, which provide additional protection for the product and associated data transfer, must be implemented by the operators themselves in line with their security standards.</p> <p><b>Device-specific IT security</b></p> <p>The device offers specific functions to support protective measures by the operator. These functions can be configured by the user and guarantee greater in-operation safety if used correctly. The user role can be changed with an access code (applies to operation via Bluetooth® wireless technology or FieldCare, DeviceCare, Asset Management Tools (e.g. AMS, PDM)).</p> <p><i>Access via Bluetooth® wireless technology</i></p> <p>Secure signal transmission via Bluetooth® wireless technology uses an encryption method tested by the Fraunhofer Institute.</p> <ul style="list-style-type: none"><li>■ Without the SmartBlue app, the device is not visible via Bluetooth® wireless technology.</li><li>■ Only one point-to-point connection is established between the device and a smartphone or tablet.</li><li>■ The Bluetooth® interface can be disabled via SmartBlue or an operating tool via digital communication.</li></ul> |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Input

|                   |                                                                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Measured variable | <p>The measured variable is the distance between the reference point and the product surface.</p> <p>The level is calculated based on <b>E</b>, the empty distance entered.</p>                |
| Measuring range   | <p>The measuring range starts at the point where the beam hits the tank floor. Levels below this point cannot be measured, particularly in the case of spherical bases or conical outlets.</p> |

Maximum measuring range

The maximum measuring range depends on the antenna size.

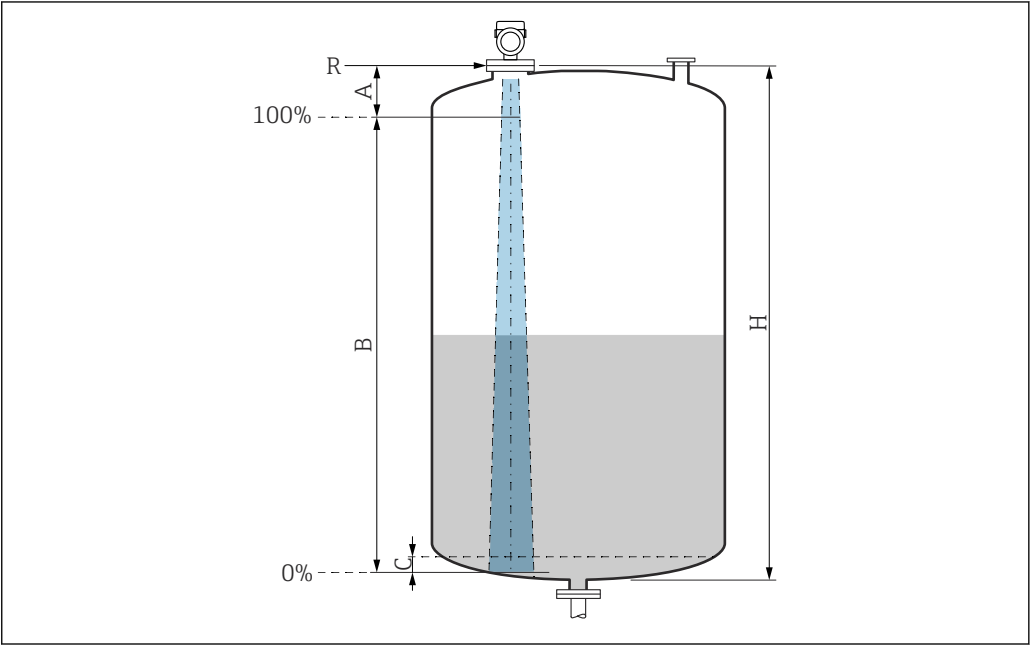
| Antenna        | Maximum measuring range |
|----------------|-------------------------|
| 40 mm (1.5 in) | 20 m (65.6 ft)          |
| 80 mm (3 in)   | 30 m (98.4 ft)          |

Usable measuring range

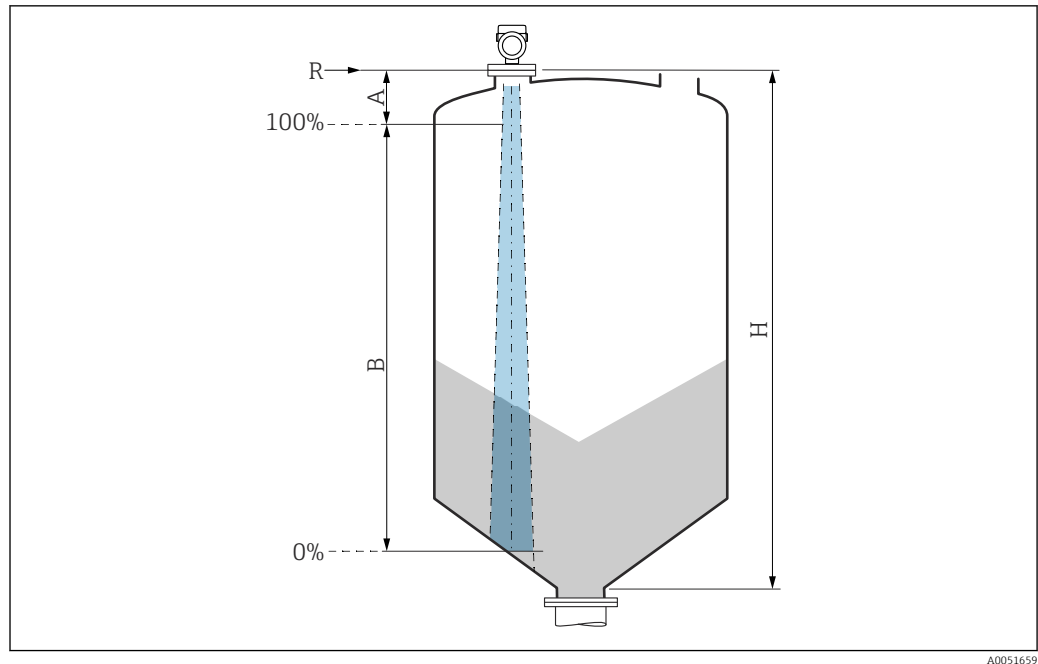
The usable measuring range depends on the antenna size, the medium's reflective properties, the installation position and any possible interference reflections.

In principle, measurement is possible up to the tip of the antenna.

Depending on the position of the product (angle of repose for solids) and to avoid any material damage from corrosive or aggressive media or deposit buildup on the antenna, the end of the measuring range should be selected 10 mm (0.4 in) before the tip of the antenna.



- A Antenna tip + 10 mm (0.4 in)
- B Usable measuring range
- C 50 to 80 mm (1.97 to 3.15 in); Medium  $\epsilon_r \leq 2$
- H Vessel height
- R Reference point of the measurement, varies depending on the antenna system (see section on Mechanical construction)



- A** Antenna tip + 10 mm (0.4 in)  
**B** Usable measuring range  
**H** Vessel height  
**R** Reference point of the measurement, varies depending on the antenna system (see section on Mechanical construction)

In the case of media with a low dielectric constant  $\epsilon_r < 2$ , the tank bottom may be visible through the medium when levels are very low (less than level C). Reduced accuracy must be expected in this range. If this is not acceptable, the zero point should be positioned at a distance C above the tank bottom for these applications (see figure).

The media groups and the possible measuring range are described as a function of the application and media group in the following section. If the relative permittivity of the medium is not known, to ensure a reliable measurement assume the medium belongs to group B.

#### Media groups

- **A** ( $\epsilon_r$  1.4 to 1.9)  
Non-conductive liquids, e.g. liquefied gas
- **B** ( $\epsilon_r$  1.9 to 4)  
Non-conductive liquids, e.g. gasoline, oil, toluene, etc.
- **C** ( $\epsilon_r$  4 to 10)  
e.g. concentrated acid, organic solvents, ester, aniline, etc.
- **D** ( $\epsilon_r > 10$ )  
Conductive liquids, aqueous solutions, diluted acids, bases and alcohol



For the relative permittivity values ( $\epsilon_r$  values) of many media commonly used in industry, please refer to:

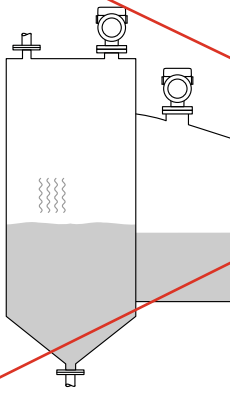
- Relative permittivity ( $\epsilon_r$  value), Compendium CP01076F
- The Endress+Hauser "DC Values app" (available for Android and iOS)

#### Measurement in storage vessel

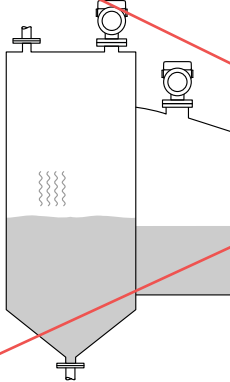
##### Storage vessel - measuring conditions

Calm medium surface (e.g. bottom filling, filling via immersion tube or rare filling from above)

40 mm (1.5 in) antenna in storage vessel

|                                                                                   | Media group                  | Measuring range |
|-----------------------------------------------------------------------------------|------------------------------|-----------------|
|  | A ( $\epsilon_r$ 1.4 to 1.9) | 10 m (33 ft)    |
|                                                                                   | B ( $\epsilon_r$ 1.9 to 4)   | 20 m (65.6 ft)  |
|                                                                                   | C ( $\epsilon_r$ 4 to 10)    | 20 m (65.6 ft)  |
|                                                                                   | D ( $\epsilon_r > 10$ )      | 20 m (65.6 ft)  |

80 mm (3 in) antenna in storage vessel

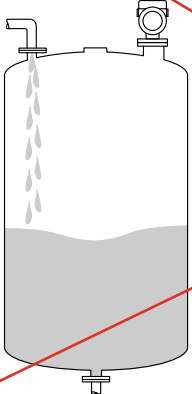
|                                                                                    | Media group                  | Measuring range |
|------------------------------------------------------------------------------------|------------------------------|-----------------|
|  | A ( $\epsilon_r$ 1.4 to 1.9) | 12 m (39 ft)    |
|                                                                                    | B ( $\epsilon_r$ 1.9 to 4)   | 23 m (75 ft)    |
|                                                                                    | C ( $\epsilon_r$ 4 to 10)    | 30 m (98 ft)    |
|                                                                                    | D ( $\epsilon_r > 10$ )      | 30 m (98 ft)    |

Measurement in buffer vessel

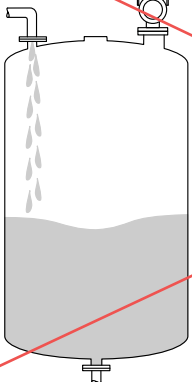
**Buffer vessel - measuring conditions**

Moving medium surface (e.g. permanent free filling from above, mixing jets)

40 mm (1.5 in) antenna in buffer vessel

|                                                                                     | Media group                  | Measuring range |
|-------------------------------------------------------------------------------------|------------------------------|-----------------|
|  | A ( $\epsilon_r$ 1.4 to 1.9) | 7 m (23 ft)     |
|                                                                                     | B ( $\epsilon_r$ 1.9 to 4)   | 13 m (43 ft)    |
|                                                                                     | C ( $\epsilon_r$ 4 to 10)    | 20 m (65.6 ft)  |
|                                                                                     | D ( $\epsilon_r > 10$ )      | 20 m (65.6 ft)  |

80 mm (3 in) antenna in buffer vessel

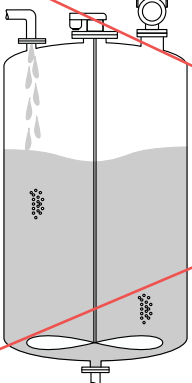
|                                                                                   | Media group                  | Measuring range |
|-----------------------------------------------------------------------------------|------------------------------|-----------------|
|  | A ( $\epsilon_r$ 1.4 to 1.9) | 7.5 m (25 ft)   |
|                                                                                   | B ( $\epsilon_r$ 1.9 to 4)   | 15 m (49 ft)    |
|                                                                                   | C ( $\epsilon_r$ 4 to 10)    | 28 m (92 ft)    |
|                                                                                   | D ( $\epsilon_r > 10$ )      | 30 m (98 ft)    |

Measurement in vessel with agitator

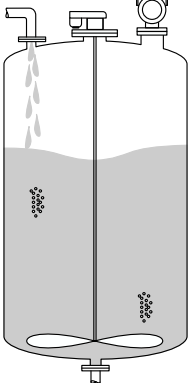
**Vessel with agitator - measuring conditions**

Turbulent medium surface (e.g. from filling from above, stirrers and baffles)

40 mm (1.5 in) antenna in vessel with agitator

|                                                                                    | Media group                  | Measuring range |
|------------------------------------------------------------------------------------|------------------------------|-----------------|
|  | A ( $\epsilon_r$ 1.4 to 1.9) | 4 m (13 ft)     |
|                                                                                    | B ( $\epsilon_r$ 1.9 to 4)   | 5 m (16.4 ft)   |
|                                                                                    | C ( $\epsilon_r$ 4 to 10)    | 13 m (43 ft)    |
|                                                                                    | D ( $\epsilon_r > 10$ )      | 20 m (65.6 ft)  |

80 mm (3 in) antenna in vessel with agitator

|                                                                                     | Media group                  | Measuring range |
|-------------------------------------------------------------------------------------|------------------------------|-----------------|
|  | A ( $\epsilon_r$ 1.4 to 1.9) | 4 m (13 ft)     |
|                                                                                     | B ( $\epsilon_r$ 1.9 to 4)   | 7 m (23 ft)     |
|                                                                                     | C ( $\epsilon_r$ 4 to 10)    | 15 m (49 ft)    |
|                                                                                     | D ( $\epsilon_r > 10$ )      | 25 m (82 ft)    |

Operating frequency

approx. 80 GHz

Up to eight devices can be installed in a tank without the devices mutually influencing one another.

**Transmission power**

- Peak power: <1.5 mW
- Average output power: <70 µW

## Output

**Output signal**

- 4 to 20 mA with superimposed digital communication protocol HART, 2-wire
- The current output offers a choice of three different operating modes:
  - 4 to 20.5 mA
  - NAMUR NE 43: 3.8 to 20.5 mA (factory setting)
  - US mode: 3.9 to 20.5 mA

**Signal on alarm for devices with current output****Current output**

Signal on alarm in accordance with NAMUR recommendation NE 43.

- Max. alarm: can be set from 21.5 to 23 mA
- Min. alarm: < 3.6 mA (factory setting)

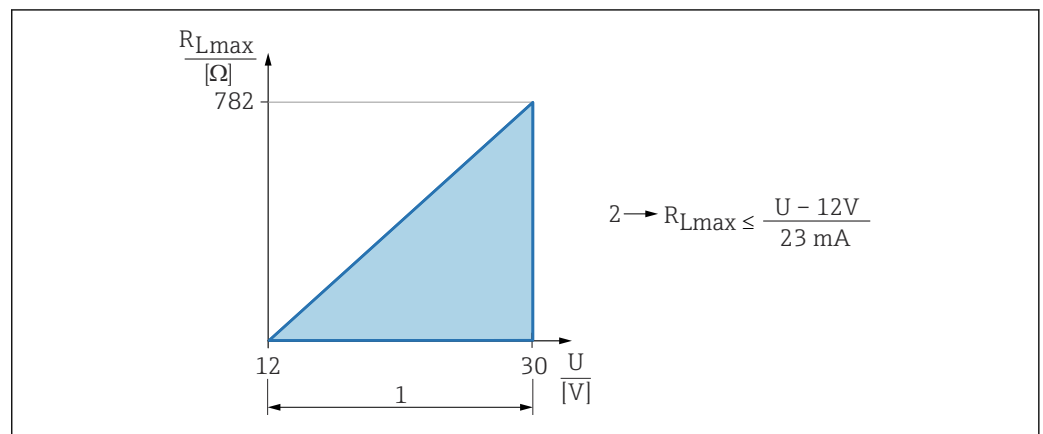
**Operating tool via digital communication**

Status signal (according to NAMUR Recommendation NE 107):

Plain text display

**Load**

In order to guarantee sufficient terminal voltage, a maximum load resistance  $R_L$  (including line resistance) must not be exceeded, depending on the supply voltage  $U$  of the supply unit.



A0052602

- 1 Power supply 12 to 30 V  
 2  $R_{Lmax}$  maximum load resistance  
 U Supply voltage

If load is too great:

- Failure current is indicated and error message is displayed (indication: MIN alarm current)
- Periodic checking to establish if it is possible to quit fault state



Operation via handheld terminal or PC with operating program: take minimum communication resistor of 250 Ω into consideration.

**Damping**

Damping affects all continuous outputs.  
 Factory setting: 0 s (can be set from 0 to 999 s)

**Ex connection data**

See the separate technical documentation (Safety Instructions (XA)) on [www.endress.com/download](http://www.endress.com/download).

**Linearization**

The linearization function of the device allows the conversion of the measured value into any unit of length, weight, flow or volume.

**Pre-programmed linearization curves**

Linearization tables for calculating the volume in the following vessels are pre-programmed into the device:

- Pyramid bottom
- Conical bottom
- Angled bottom
- Horizontal cylinder
- Sphere

Linearization tables for calculating the flow rate are pre-programmed into the device and include the following:

- Flumes
  - Khafagi Venturi flume
  - Venturi flume
  - Parshall flume
  - Palmer Bowlus flume
  - Trapezoidal flume (ISO 4359)
  - Rectangular flume (ISO 4359)
  - U-shaped flume (ISO 4359)
- Weirs
  - Trapezoidal weir
  - Rectangular broad-crested weir (ISO 3846)
  - Rectangular thin-plate weir (ISO 1438)
  - V-notch thin-plate weir (ISO 1438)
- Standard formula

Other linearization tables of up to 32 value pairs can be entered manually.



For further information on flow measurement over open channels and weirs, see SD03445F.

|                  |                                                                                           |
|------------------|-------------------------------------------------------------------------------------------|
| <b>Totalizer</b> | The device offers a totalizer which adds up the flow rate. The totalizer cannot be reset. |
|------------------|-------------------------------------------------------------------------------------------|

|                               |                                                                                                                                                                                                       |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Protocol-specific data</b> | <p><b>Manufacturer ID:</b><br/>17(0x0011)</p> <p><b>Device type ID:</b><br/>0x11DE</p> <p><b>Device revision:</b><br/>2</p> <p><b>HART specification:</b><br/>7.6</p> <p><b>DD version:</b><br/>1</p> |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Device description files (DTM, DD)**

Information and files at:

- [www.endress.com](http://www.endress.com)  
On the product page for the device: Documents/Software → Device drivers
- [www.fieldcommgroup.org](http://www.fieldcommgroup.org)

**HART load:**

Min. 250 Ω

The following measured values are assigned to the device variables at the factory:

| Device variable                     | Measured value          |
|-------------------------------------|-------------------------|
| Primary variable (PV) <sup>1)</sup> | Level linearized        |
| Secondary variable (SV)             | Distance                |
| Tertiary variable (TV)              | Absolute echo amplitude |
| Quaternary variable (QV)            | Relative echo amplitude |

1) The PV is always applied to the current output.

**Choice of HART device variables**

- Level linearized
- Distance
- Electronics temperature
- Sensor temperature
- Absolute echo amplitude
- Relative echo amplitude
- Area of incoupling
- Percent of range
- Loop current
- Flow
- Totalizer value
- Not used

**Supported functions**

- Burst mode
- Additional transmitter status
- Device locking

**Wireless HART data**

**Minimum start-up voltage:**  
12 V

**Start-up current:**  
< 3.6 mA

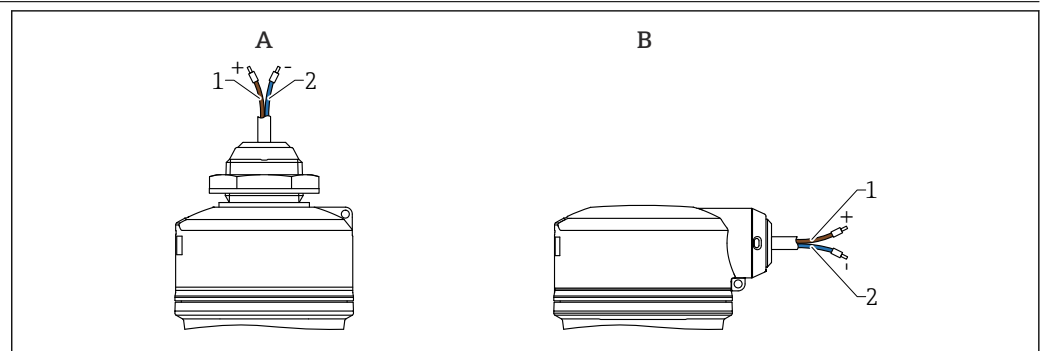
**Starting time:**  
< 15 s

**Minimum operating voltage:**  
12 V

**Multidrop current:**  
4 mA

**Time to establish connection:**  
< 30 s

## Power supply

**Cable assignment**

A0055191

**3 Cable assignment**

**A** Cable entry from above

**B** Side cable entry

**1** Plus, brown wire

**2** Minus, blue wire

**Supply voltage**

12 to 30 V DC on a DC power unit

**i** The power unit must be safety-approved (e.g. PELV, SELV, Class 2) and must comply with the relevant protocol specifications.

Protective circuits against reverse polarity, HF influences and overvoltage peaks are installed.

**Power consumption**

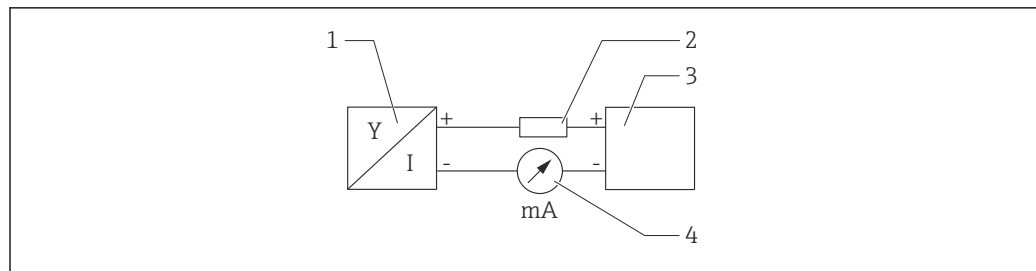
- Non-hazardous area: To meet device safety specifications according to the IEC 61010 standard, the installation must ensure that the maximum current is limited to 500 mA.
- Hazardous area: The maximum current is restricted to  $I_i = 100$  mA by the transmitter power supply unit when the measuring instrument is used in an intrinsically safe circuit (Ex ia).

**Potential equalization**

No special measures for potential equalization are required.

**Connecting the device****Function diagram of 4 to 20 mA HART**

Connection of the device with HART communication, power source and 4 to 20 mA indicator



A0028908

4 Function diagram of HART connection

- 1 Device with HART communication
- 2 HART resistor
- 3 Power supply
- 4 Multimeter or ammeter

**i** The HART communication resistor of 250  $\Omega$  in the signal line is always necessary in the case of a low-impedance power supply.

**The voltage drop to be taken into account is:**  
Max. 6 V for 250  $\Omega$  communication resistor

**Function diagram of HART device, connection with RIA15, display only without operation, without communication resistor**

**i** The RIA15 remote display can be ordered together with the device.

**i** Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

**Terminal assignment RIA15**

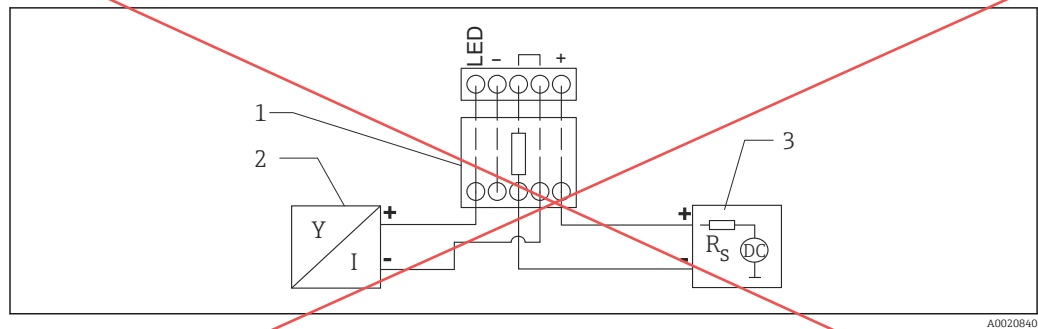
- +  
Positive connection, current measurement
- -  
Negative connection, current measurement (without backlighting)
- LED  
Negative connection, current measurement (with backlighting)
- $\equiv$   
Functional grounding: Terminal in housing

**i** The RIA15 process indicator is loop-powered and does not require any external power supply.

**The voltage drop to be taken into account is:**

- $\leq 1$  V in the standard version with 4 to 20 mA communication
- $\leq 1.9$  V with HART communication
- and an additional 2.9 V if display light is used

Connection of the HART communication resistor module, RIA15 with backlighting



8 Function diagram of HART device, RIA15 with light, HART communication resistor module

- 1 HART communication resistor module  
 2 Device with HART communication  
 3 Current supply

### Cable specification

Unshielded cable, wire cross-section 0.5 mm<sup>2</sup>

- Resistant to UV and weather conditions as per ISO 4892-2
- Flame resistance according to IEC 60332-1-2

As per IEC 60079-11 section 9.4.4, the cable is designed for a tensile strength of 30 N (6.74 lbf) (over a period of 1 h).

The device is available in cable lengths of 5 m (16 ft), 10 m (32 ft), 15 m (49 ft), 20 m (65 ft), 30 m (98 ft) and 50 m (164 ft).

User-defined lengths up to total length of 300 m (980 ft) are possible in increments of one meter (order option "1") or one foot (order "2").

For devices with marine approval:

- Only available with a length of 10 m (32 ft) and "user-defined"
- Halogen-free as per IEC 60754-1
- No development of corrosive fire gases in accordance with IEC 60754-2
- Low flue gas density in accordance with IEC 61034-2

### Overvoltage protection

The device satisfies the IEC/DIN EN 61326-1 product standard (Table 2 Industrial environment). Depending on the type of connection (DC power supply, input line, output line), different test levels are used to prevent transient overvoltages (IEC/DIN EN 61000-4-5 Surge) in accordance with IEC/DIN EN 61326-1: Test level for DC power supply lines and IO lines: 1 000 V wire to ground.

Devices for the "protection by enclosure" explosion protection are equipped with an integrated overvoltage protection system.

#### Overvoltage category

In accordance with IEC/DIN EN 61010-1, the device is intended for use in networks with overvoltage protection category II.

## Performance characteristics

### Reference operating conditions

- As per IEC 62828-1/IEC 62828-4
- Ambient temperature  $T_A$  = constant, in the range of +21 to +33 °C (+70 to +91 °F)
- Humidity  $\phi$  = constant, in the range of: 5 to 80 % RH  $\pm$  5 %
- Atmospheric pressure  $p_U$  = constant, in the range of: 860 to 1 060 mbar (12.47 to 15.37 psi)
- Load with HART: 250  $\Omega$
- Supply voltage: DC 24 V  $\pm$  3 V
- Reflector: metal plate with diameter  $\geq$  1 m (40 in)
- No major interference echoes inside the signal beam

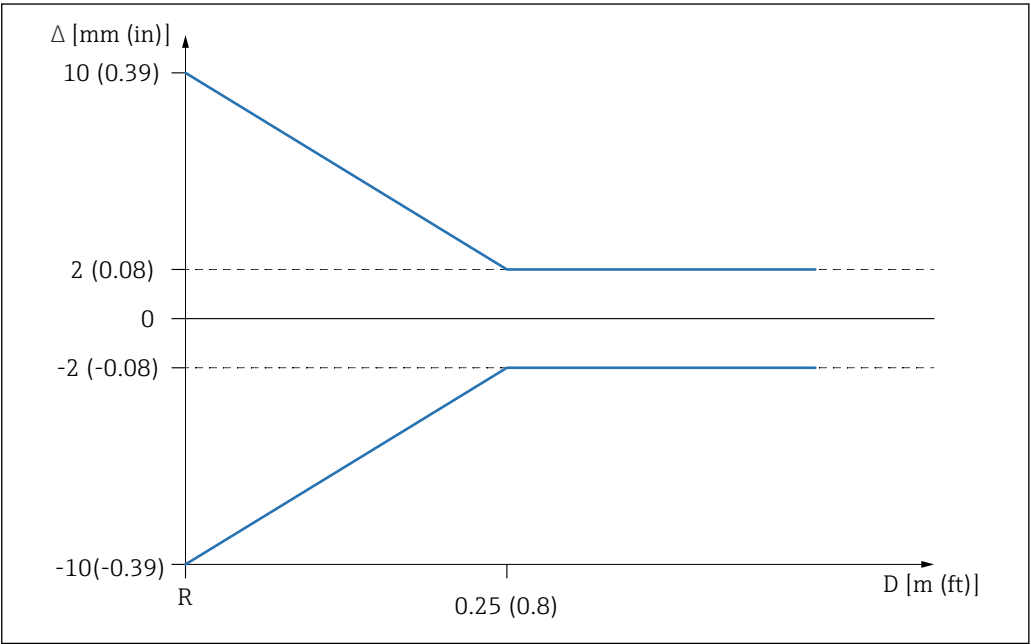
### Response time

- HART: acyclic: min. 330 ms, typically 590 ms (depending on commands and number of preambles)
- HART: cyclic (burst): min. 160 ms, typically 350 ms (depending on commands and number of preambles)

|            |                                                        |
|------------|--------------------------------------------------------|
| Resolution | Current output: < 1 $\mu$ A<br>Digital: 1 mm (0.04 in) |
|------------|--------------------------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maximum measurement error | <b>Reference accuracy</b><br><b>Accuracy</b><br>The accuracy is the sum of the non-linearity, non-repeatability and hysteresis.<br>For liquids: <ul style="list-style-type: none"><li>■ Measuring distance up to 0.25 m (0.82 ft): max. <math>\pm 10</math> mm (<math>\pm 0.39</math> in)</li><li>■ Measuring distance &gt; 0.25 m (0.82 ft): <math>\pm 2</math> mm (<math>\pm 0.08</math> in)</li></ul> For solids: <ul style="list-style-type: none"><li>■ Measuring distance up to 0.8 m (2.6 ft): max. <math>\pm 20</math> mm (<math>\pm 0.79</math> in)</li><li>■ Measuring distance &gt; 0.8 m (2.6 ft): <math>\pm 4</math> mm (<math>\pm 0.16</math> in)</li></ul> <b>Non-repeatability</b><br>Non-repeatability is already included in the accuracy.<br>$\leq 1$ mm (0.04 in)<br> If conditions deviate from the reference operating conditions, the offset/zero point that results from the installation conditions can be up to $\pm 4$ mm ( $\pm 0.16$ in). This additional offset/zero point can be eliminated by entering a correction ( <b>Level correction</b> parameter) during commissioning. |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

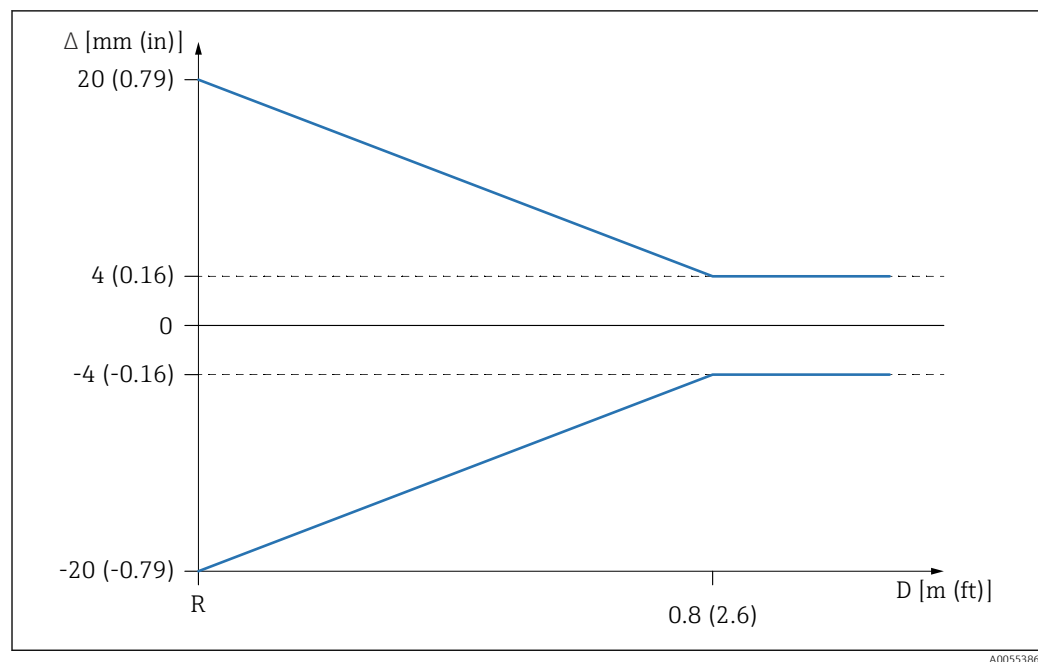
**Differing values in near-range applications for liquids**



 9 Maximum measurement error in near-range applications

- $\Delta$  Maximum measurement error
- $R$  Reference point of the distance measurement
- $D$  Distance from reference point of antenna

## Differing values in near-range applications for solids



10 Maximum measurement error in near-range applications

$\Delta$  Maximum measurement error

$R$  Reference point of the distance measurement

$D$  Distance from reference point of antenna

### Influence of ambient temperature

The output changes due to the effect of the ambient temperature with respect to the reference temperature.

The measurements are performed according to IEC 61298-3 / IEC 60770-1

#### Digital output (HART)

Average  $T_C = \pm 2 \text{ mm } (\pm 0.08 \text{ in}) / 10 \text{ K}$

#### Analog (current output)

■ Zero point (4 mA): average  $T_C = 0.02 \% / 10 \text{ K}$

■ Span (20 mA): average  $T_C = 0.05 \% / 10 \text{ K}$

### Reaction time

According to IEC 61298-2 / IEC 60770-1, the step response time is the time following an abrupt change in the input signal up until the changed output signal has adopted 90 % of the steady-state value for the first time.

The response time can be configured.

The following step response times apply (in accordance with IEC 61298-2/IEC 60770-1) when damping is switched off:

- Measuring rate  $\leq 250 \text{ ms}$  at operating voltage 24 V
- Step response time  $< 1 \text{ s}$

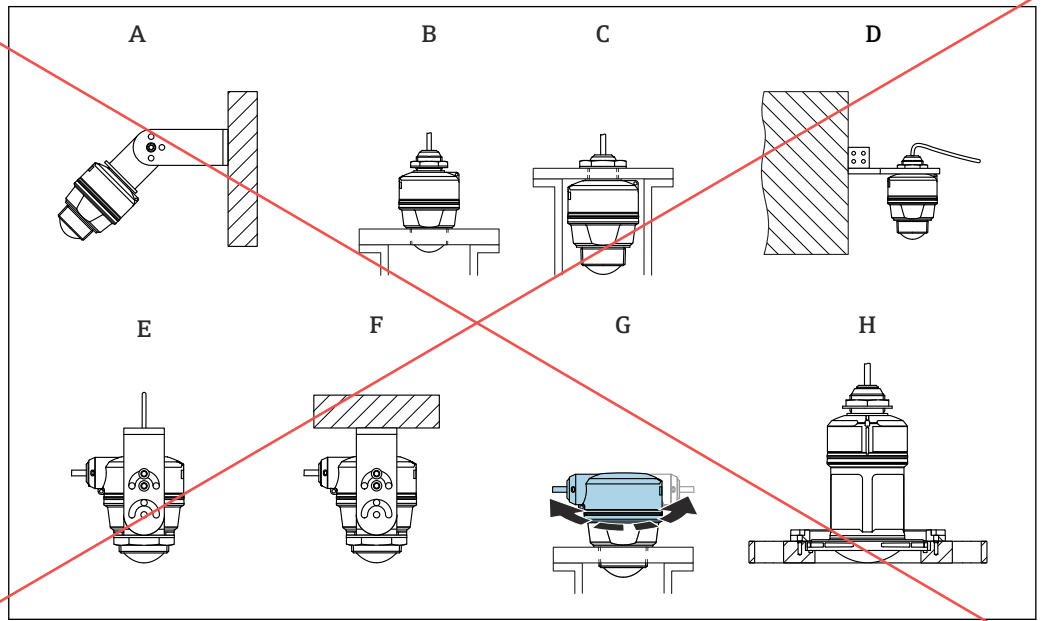
### Warm-up time

The warm-up time (in accordance with IEC 62828-4) indicates the time required for the device to reach its maximum accuracy or performance after the supply voltage is energized.

Warm-up time:  $\leq 15 \text{ s}$

## Installation

### Installation types



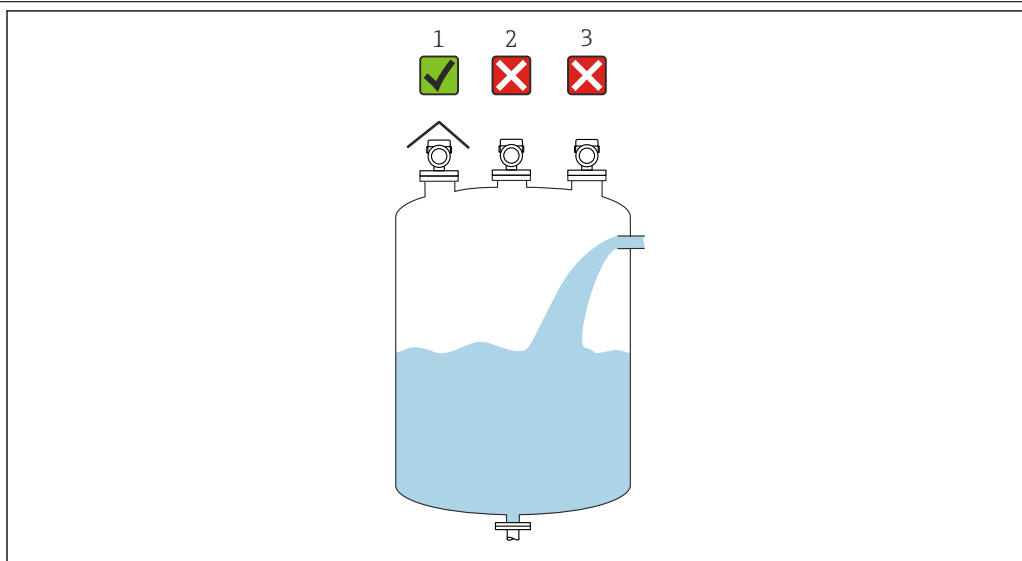
#### 11 Wall or ceiling mount

- A Wall mounting adjustable
- B Tightened at antenna end process connection
- C Tightened at cable entry from above process connection
- D Wall mounting with cable entry from above process connection
- E Rope mounting with cable entry at the side
- F Ceiling mounting with cable entry at the side
- G Cable entry at the side, top housing section can be rotated
- H Mounting with UNI slip-on flange



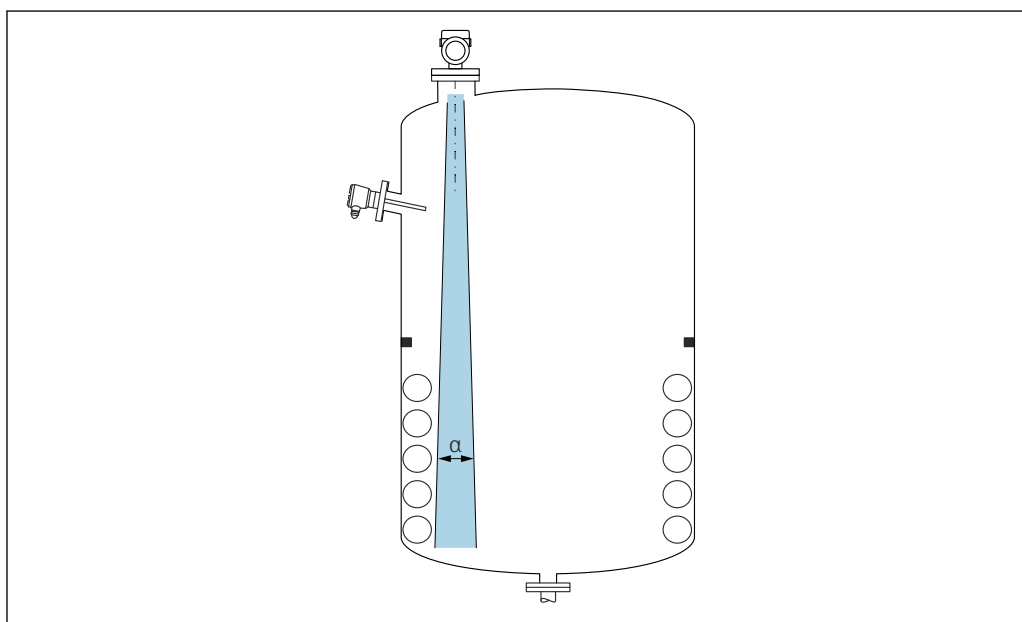
#### Please note the following:

- The sensor cables are not designed as supporting cables. Do not use them for suspension purposes.
- For rope mounting, the rope must be provided by the customer.
- Always operate the device in a vertical position in free-space applications.
- For devices with side cable outlet and 80 mm antenna, installation is only possible with a UNI slip-on flange.

**Mounting location**

A0055811

- 1 Use of a weather protection cover; protection from direct sunlight or rain
- 2 Installation not centered: Interferences can lead to incorrect signal analysis
- 3 Do not install above the filling curtain

**Orientation****Internal vessel fittings**

A0031777

Avoid internal fittings (level switches, temperature sensors, struts, vacuum rings, heating coils, baffles etc.) inside the signal beam. Pay attention to the beam angle  $\alpha$ .

**Vertical alignment of antenna axis**

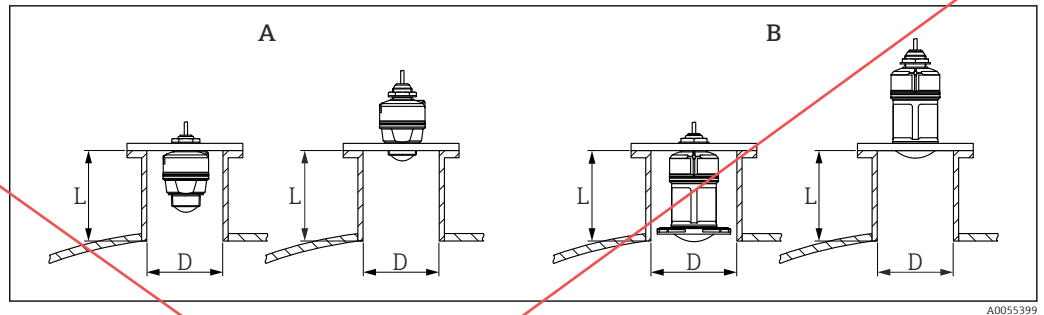
Align the antenna so that it is perpendicular to the product surface.



The maximum reach of the antenna can be reduced, or additional interference signals can occur, if the antenna is not installed perpendicular to the product.

**Installation instructions**

To ensure optimum measurement, the antenna must protrude from the nozzle. The interior of the nozzle must be smooth and must not contain any edges or welded joints. If possible, round the nozzle edge.



12 Nozzle installation

- A 40 mm (1.5 in) antenna  
B 80 mm (3 in) antenna

The maximum nozzle length **L** depends on the nozzle diameter **D**.

Please note the limits for the diameter and length of the nozzle.

**40 mm (1.5 in) antenna, installation outside nozzle**

- D: min. 40 mm (1.5 in)
- L: max.  $(D - 30 \text{ mm (1.2 in)}) \times 7.5$

**40 mm (1.5 in) antenna, installation inside nozzle**

- D: min. 80 mm (3 in)
- L: max.  $100 \text{ mm (3.94 in)} + (D - 30 \text{ mm (1.2 in)}) \times 7.5$

**80 mm (3 in) antenna, installation inside nozzle**

- D: min. 120 mm (4.72 in)
- L: max.  $140 \text{ mm (5.51 in)} + (D - 50 \text{ mm (2 in)}) \times 12$

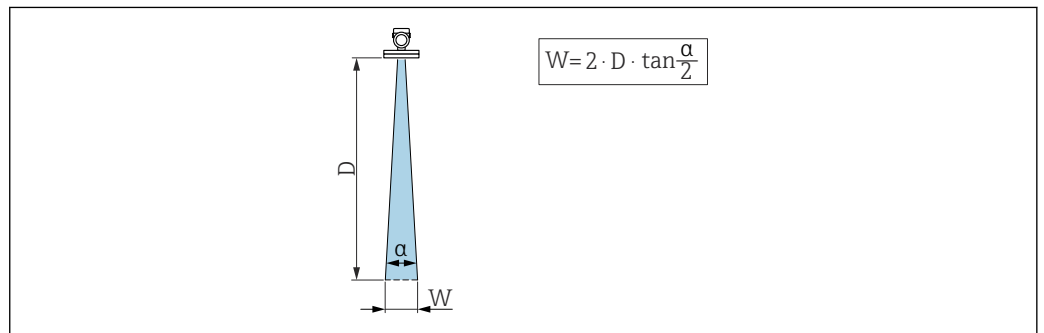
**80 mm (3 in) antenna, installation outside nozzle**

- D: min. 80 mm (3 in)
- L: max.  $(D - 50 \text{ mm (2 in)}) \times 12$

## Beam angle

## Calculation

The beam angle is defined as the angle  $\alpha$  at which the energy density of the radar waves reaches half the value of the maximum energy density (3dB width). Microwaves are also emitted outside the signal beam and can be reflected off interfering installations.

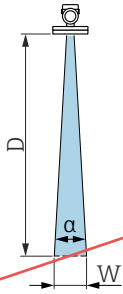


13 Relationship between beam angle  $\alpha$ , distance **D** and beamwidth diameter **W**

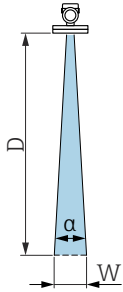


The beamwidth diameter **W** depends on the beam angle  $\alpha$  and the distance **D**.

**Antenna 40 mm (1.5 in),  $\alpha = 8^\circ$** 

| $W = D \times 0.14$                                                               | D            | W                |
|-----------------------------------------------------------------------------------|--------------|------------------|
|  | 5 m (16 ft)  | 0.70 m (2.29 ft) |
|                                                                                   | 10 m (33 ft) | 1.40 m (4.58 ft) |
|                                                                                   | 15 m (49 ft) | 2.09 m (6.87 ft) |
|                                                                                   | 20 m (66 ft) | 2.79 m (9.16 ft) |

**Antenna 80 mm (3 in),  $\alpha = 4^\circ$** 

| $W = D \times 0.07$                                                                | D            | W                |
|------------------------------------------------------------------------------------|--------------|------------------|
|  | 5 m (16 ft)  | 0.35 m (1.15 ft) |
|                                                                                    | 10 m (33 ft) | 0.70 m (2.30 ft) |
|                                                                                    | 15 m (49 ft) | 1.05 m (3.45 ft) |
|                                                                                    | 20 m (66 ft) | 1.40 m (4.59 ft) |
|                                                                                    | 25 m (82 ft) | 1.75 m (5.74 ft) |
|                                                                                    | 30 m (98 ft) | 2.10 m (6.89 ft) |

**Special installation instructions****External measurement through plastic cover or dielectric windows**

- Dielectric constant of medium:  $\epsilon_r \geq 10$
- The distance from the tip of the antenna to the tank should be approx. 100 mm (4 in).
- Avoid installation positions where condensate or buildup can form between the antenna and the vessel
- In the case of outdoor installations, ensure that the area between the antenna and the tank is protected from the weather
- Do not install any fittings or attachments between the antenna and the tank that could reflect the signal

The thickness of the tank ceiling or the dielectric window depends on the  $\epsilon_r$  of the material.

The material thickness can be a full multiple of the optimum thickness (table); it is important to note, however, that the microwave transparency decreases significantly with increasing material thickness.

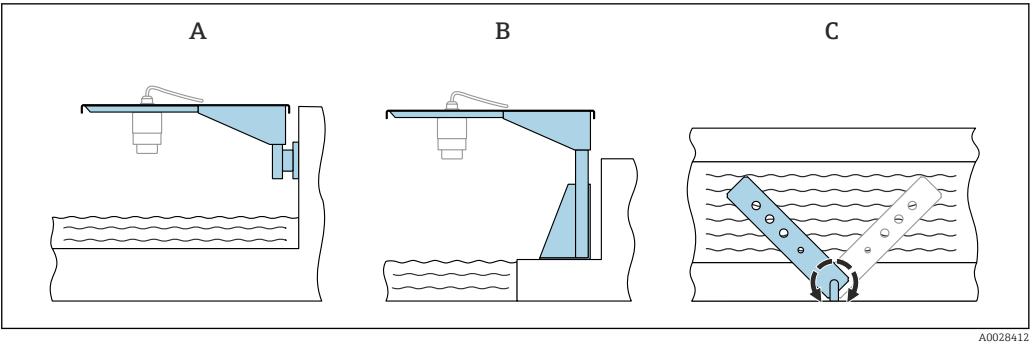
*Optimum material thickness*

| Material                  | Optimum material thickness |
|---------------------------|----------------------------|
| PE; $\epsilon_r$ 2.3      | 1.25 mm (0.049 in)         |
| PTFE; $\epsilon_r$ 2.1    | 1.30 mm (0.051 in)         |
| PP; $\epsilon_r$ 2.3      | 1.25 mm (0.049 in)         |
| Perspex; $\epsilon_r$ 3.1 | 1.10 mm (0.043 in)         |

**Weather protection cover**

A weather protective cover is recommended for outdoor use.

The weather protective cover can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".



18 Cantilever installation, with pivot

- A Cantilever with wall bracket (side view)
- B Cantilever with mounting frame (side view)
- C Cantilever can be turned, e.g. in order to position the device over the center of the flume (top view)

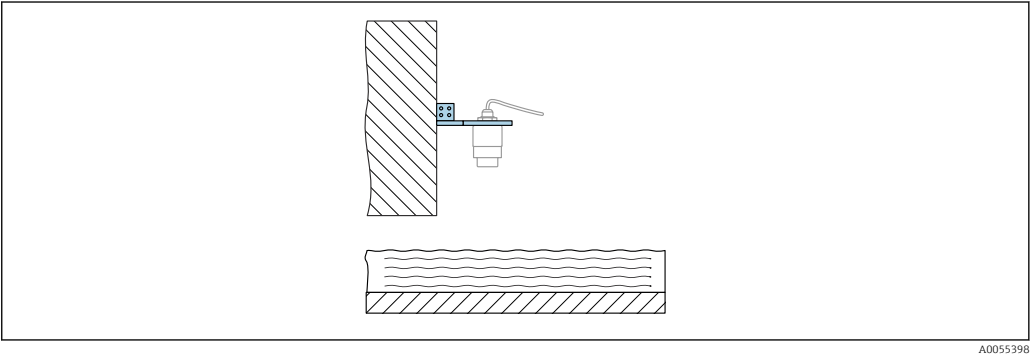
**NOTICE**

**There is no conductive connection between the mounting bracket and transmitter housing.**  
Electrostatic charging possible.

- Integrate the mounting bracket in the local potential equalization system.

**Mounting with a pivotable mounting bracket**

The pivotable mounting bracket can be ordered as an accessory or together with the device via the product structure "Accessory enclosed".



19 Pivotable and adjustable cantilever with wall bracket (e.g. to align the device with the center of a flume)

**NOTICE**

**There is no conductive connection between the mounting bracket and transmitter housing.**  
Electrostatic charging possible.

- Integrate the mounting bracket in the local potential equalization system.

## Environment

|                                  |                                                                                                                                                                                                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ambient temperature range</b> | <p>–40 to +80 °C (–40 to +176 °F)</p> <p>If operating outdoors in strong sunlight:</p> <ul style="list-style-type: none"> <li>■ Mount the device in the shade.</li> <li>■ Avoid direct sunlight, particularly in warmer climatic regions.</li> <li>■ Use a weather protective cover.</li> </ul> |
| <b>Storage temperature</b>       | –40 to +80 °C (–40 to +176 °F)                                                                                                                                                                                                                                                                  |
| <b>Climate class</b>             | According to IEC 60068-2-38 test Z/AD (relative humidity 4 to 100 %).                                                                                                                                                                                                                           |
| <b>Operating height</b>          | Up to 5 000 m (16 404 ft) above sea level                                                                                                                                                                                                                                                       |

|                                            |                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Degree of protection</b>                | Testing according to IEC 60529 and NEMA 250: <ul style="list-style-type: none"><li>■ IP66, NEMA Type 4X</li><li>■ IP68, NEMA Type 6P (24 h at 1.83 m (6.00 ft) under water)</li></ul>                                                                                                                                                                                  |
| <b>Vibration resistance</b>                | <ul style="list-style-type: none"><li>■ Stochastic noise (random sweep) as per IEC 60068-2-64 Case 2</li><li>■ Guaranteed for 5 to 2 000 Hz: <math>1.25 \text{ (m/s}^2\text{)}^2/\text{Hz}</math>, ~ 5 g</li></ul>                                                                                                                                                     |
| <b>Electromagnetic compatibility (EMC)</b> | <ul style="list-style-type: none"><li>■ Electromagnetic compatibility as per EN 61326 series and NAMUR recommendation EMC (NE21)</li><li>■ Maximum measured error during EMC testing: &lt; 0.5 % of the span.</li></ul> <p>For more details, refer to the EU Declaration of Conformity (<a href="http://www.endress.com/downloads">www.endress.com/downloads</a>).</p> |

## Process

### Process temperature, process pressure



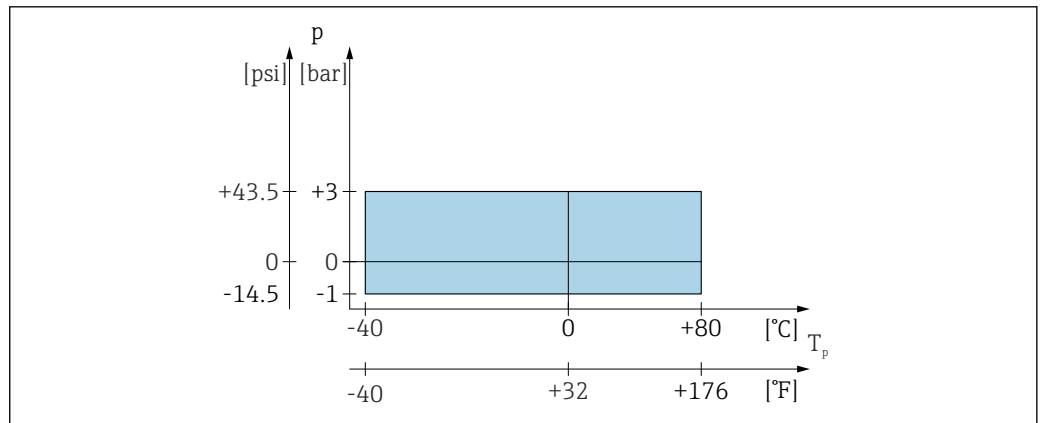
The maximum pressure for the device depends on the lowest-rated element with regard to pressure.

Components are: process connection, optional mounting parts, or accessories.

#### **WARNING**

#### **Incorrect design or use of the device may cause injury due to bursting parts!**

- ▶ Only operate the device within the specified limits for the components!
- ▶ MWP (Maximum Working Pressure): The MWP is specified on the nameplate. This value refers to a reference temperature of +20 °C (+68 °F) and may be applied to the device for an unlimited time. Observe the temperature dependency of the maximum working pressure. For flanges, refer to the following standards for the permitted pressure values at higher temperatures: EN 1092-1 (with regard to their stability/temperature property, the materials 1.4435 and 1.4404 are grouped together under EN 1092-1; the chemical composition of the two materials can be identical), ASME B16.5, JIS B2220 (the latest version of the standard applies in each case). Maximum working pressure data that deviate from this are provided in the relevant sections of the Technical Information.
- ▶ The Pressure Equipment Directive (2014/68/EU) uses the abbreviation **PS**. This corresponds to the maximum working pressure (MWP) of the device.



A0056003

20 Permitted range for process temperature and process pressure

#### **Process temperature range**

-40 to +80 °C (-40 to +176 °F)

#### **Process pressure range, 40 mm (1.5 in) antenna**

- $p_{\text{gauge}} = -1 \text{ to } 3 \text{ bar}$  (-14.5 to 43.5 psi)
- $p_{\text{abs}} < 4 \text{ bar}$  (58 psi)

#### **Process pressure range, 80 mm (3 in) antenna with UNI slip-on flange 3", 4"**

- $p_{\text{gauge}} = -1 \text{ to } 1 \text{ bar}$  (-14.5 to 14.5 psi)
- $p_{\text{abs}} < 2 \text{ bar}$  (29 psi)

#### **Process pressure range, 80 mm (3 in) antenna with UNI slip-on flange 6"**

For unpressurized applications



The pressure range may be further restricted in the case of a CRN approval.

### Relative permittivity

#### **For liquids**

- $\epsilon_r \geq 1.8$
- Contact Endress+Hauser for lower  $\epsilon_r$  values

#### **For bulk solids**

$\epsilon_r \geq 1.6$

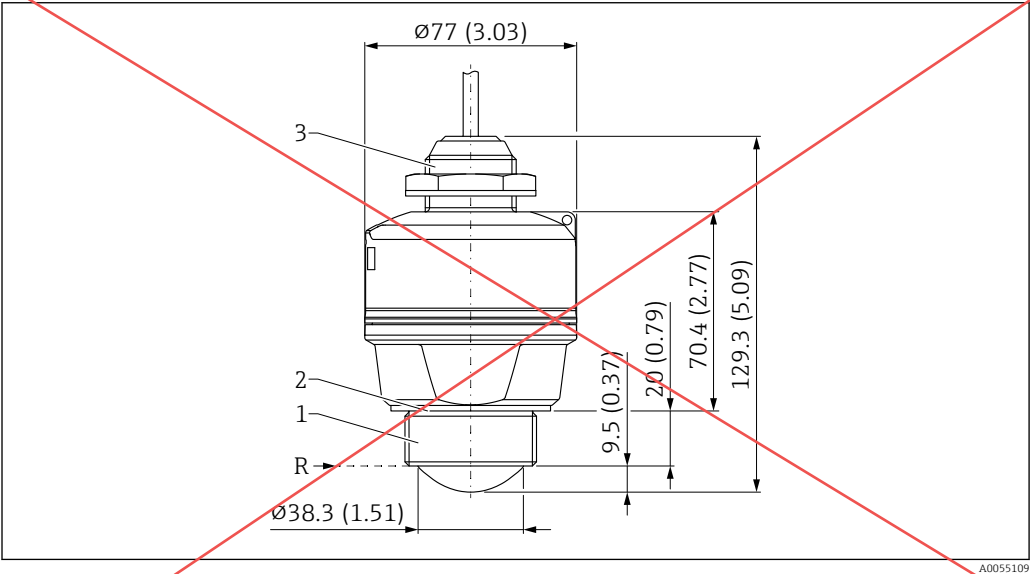
For applications with a lower relative permittivity than indicated, contact Endress+Hauser.

- i** For the relative permittivity values ( $\epsilon_r$  values) of many media commonly used in industry, please refer to:
- Relative permittivity ( $\epsilon_r$  value), Compendium CP01076F
  - The Endress+Hauser "DC Values app" (available for Android and iOS)

## Mechanical construction

### Dimensions

40 mm (1.5 in) antenna, cable entry from above

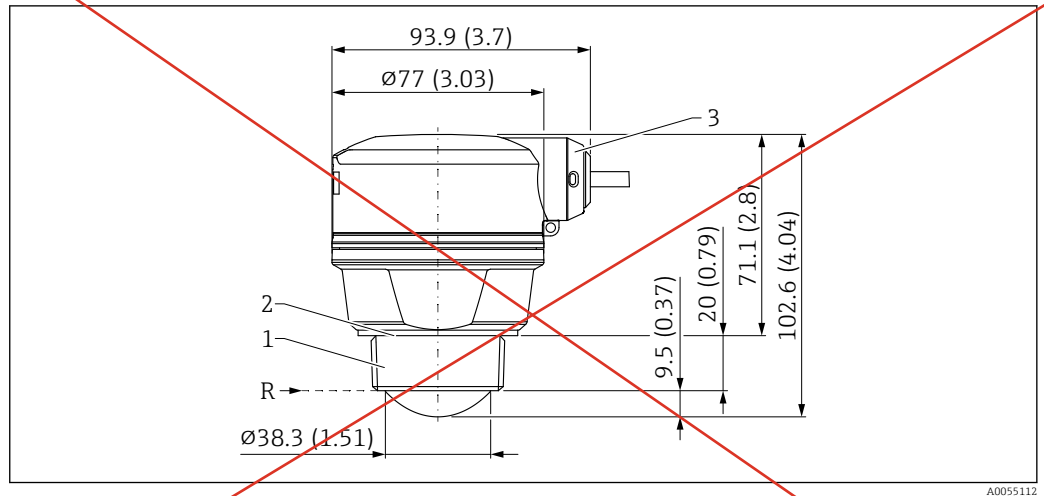


**21** Dimensions; 40 mm (1.5 in) antenna with cable entry from above. Unit of measurement mm (in)

- R Reference point of the measurement  
1 Antenna end process connection, thread  
2 EPDM seal (G 1 1/2 thread)  
3 Cable entry from above process connection

**i** The seal thickness is 2 mm (0.08 in).

### 40 mm (1.5 in) antenna, cable entry at the side

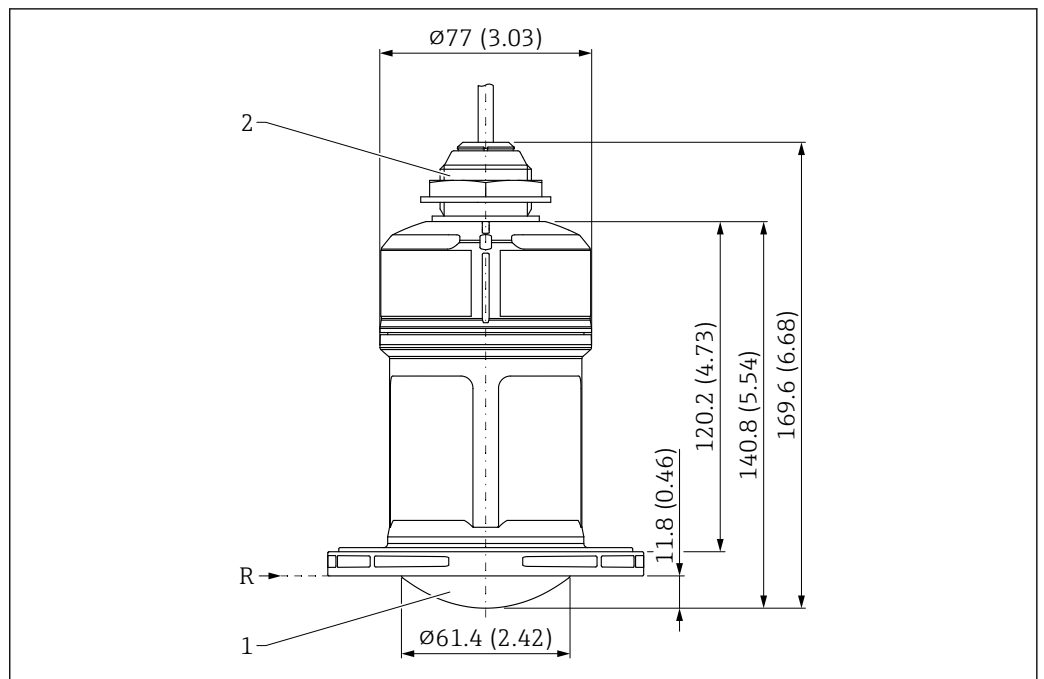


22 Dimensions; 40 mm (1.5 in) antenna with cable entry at the side. Unit of measurement mm (in)

- R Reference point of the measurement  
 1 Antenna end process connection, thread  
 2 EPDM seal (G 1 1/2 thread)  
 3 Side cable entry

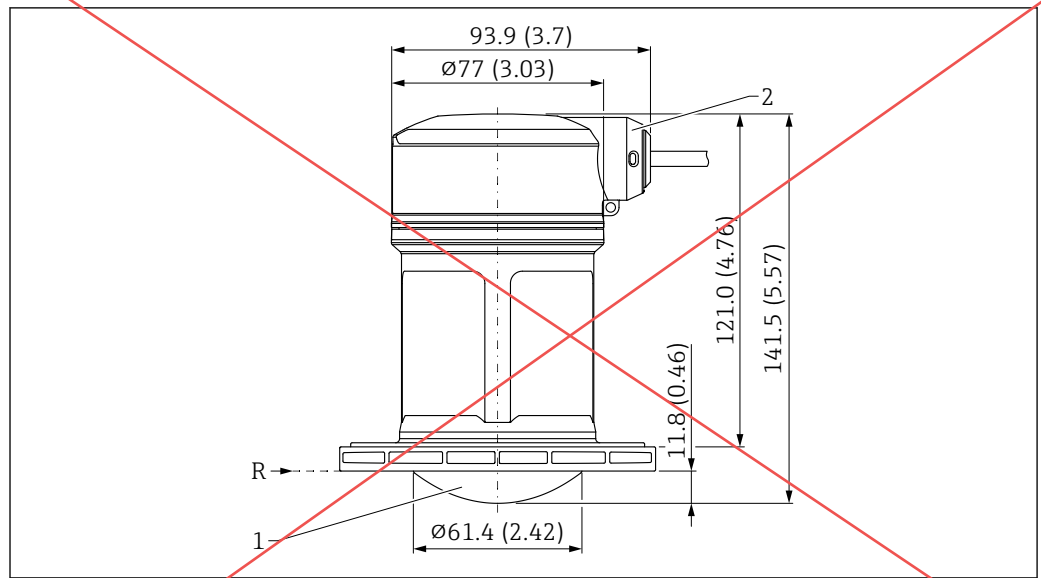
**i** The seal thickness is 2 mm (0.08 in).

### 80 mm (3 in) antenna, cable entry from above



23 Dimensions; 80 mm (3 in) antenna with cable entry from above. Unit of measurement mm (in)

- R Reference point of the measurement  
 1 Antenna end process connection, without; prepared for UNI slip-on flange  
 2 Cable entry from above process connection

**80 mm (3 in) antenna, cable entry at the side**

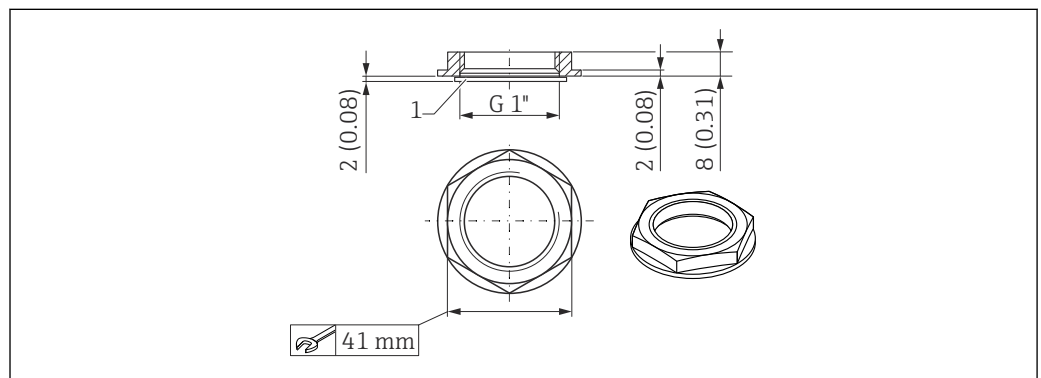
A0055110

■ 24 Dimensions; 80 mm (3 in) antenna with cable entry at the side

R Reference point of the measurement

1 Antenna end process connection, without; prepared for UNI slip-on flange

2 Side cable entry

**Counter nut, cable entry from above process connection**

A0028419

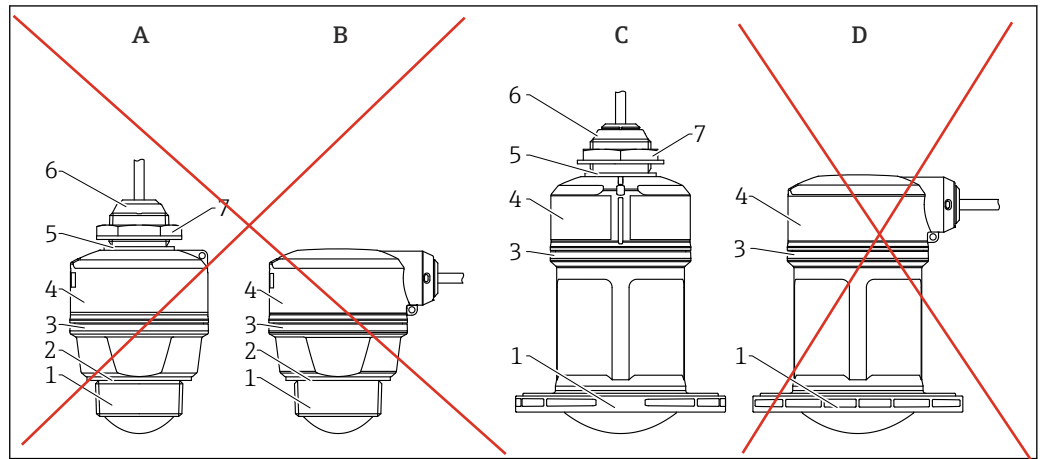
■ 25 Dimensions; counter nut, cable entry from above process connection. Unit of measurement mm (in)

1 Seal

- The counter nut with seal (EPDM) is included in the scope of delivery
- Material: PA6.6

**Weight****Weight (including 5 m (16 ft) cable)**

- Device with 40 mm (1.5 in) antenna: approx. 0.5 kg (1.1 lb)
- Device with 80 mm (3 in) antenna: approx. 0.7 kg (1.5 lb)

**Materials**

26 Device design

~~A 40 mm (1.5 in) antenna, cable entry from above~~

~~B 40 mm (1.5 in) antenna, cable entry at the side~~

~~C 80 mm (3 in) antenna, cable entry from above~~

~~D 80 mm (3 in) antenna, cable entry at the side~~

1 Antenna end process connection; PVDF

2 EPDM seal (for G 1½" thread)

3 PBT/PC design ring

4 Sensor housing/Cable entry process connection; PBT/PC (for dust ignition-proof devices: PC)

5 EPDM seal

6 Cable entry; PBT/PC (for dust ignition-proof devices: PC)

7 Counter nut; PA6.6

**Connecting cable**

Available cable length: 5 to 300 m (16 to 980 ft)

Material : PVC

For devices with marine approval: halogen-free cable (material: XLPE = connected polyethylene)

**Operability****Operating concept****Operator-oriented menu structure for user-specific tasks**

- Guidance
- Diagnostics
- Application
- System

**Fast and safe commissioning**

- Interactive wizard with graphical interface for guided commissioning in FieldCare/DeviceCare or SmartBlue app
- Menu guidance with brief descriptions of the individual parameter functions

**Integrated data memory**

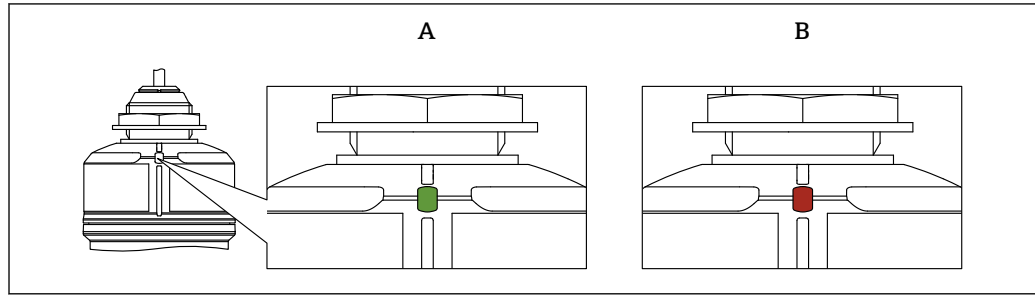
Up to 100 event messages recorded in the device

**Efficient diagnostic behavior increases measurement reliability**

- Remedial action is integrated in plain text
- Diverse simulation options

**Bluetooth® wireless technology (optional)**

- Quick and easy setup with the SmartBlue app or Field Xpert SMT70/SMT77
- No additional tools or adapters required
- Encrypted single point-to-point data transmission (tested by Fraunhofer Institute) and password-protected communication via Bluetooth® wireless technology
- The device can be retrofitted with Bluetooth® wireless technology

**LED indicator**

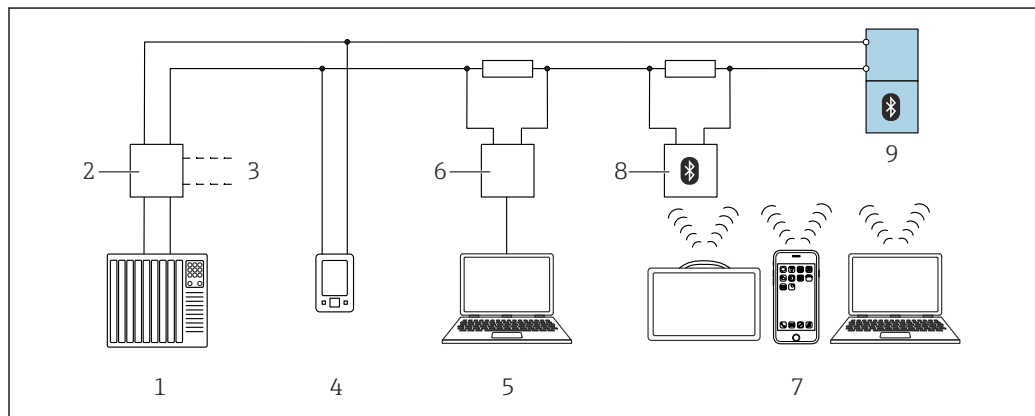
A0055146

27 LED indicator on the device

- A Green LED  
B Red LED

**Functions:**

- Displays the operating status
  - Operation (green)
  - Fault (red)
- Displays an active Bluetooth® connection (flashing)

**Remote operation****Via HART protocol or Bluetooth® wireless technology**

A0044334

28 Options for remote operation via HART protocol

- 1 PLC (programmable logic controller)
- 2 Transmitter power supply unit, e.g. RN42 (with communication resistor)
- 3 Connection for Commubox FXA195 and AMS Trex™ device communicator
- 4 AMS Trex™ device communicator
- 5 Computer with operating tool (e.g. DeviceCare/FieldCare, AMS Device View, SIMATIC PDM)
- 6 Commubox FXA195 (USB)
- 7 Field Xpert SMT70/SMT77, smartphone or computer with operating tool (e.g. DeviceCare)
- 8 Bluetooth® modem with connecting cable (e.g. VIATOR)
- 9 Transmitter

**Operation via Bluetooth® wireless technology (optional)****Prerequisite**

- Device with Bluetooth® wireless technology order option
- Smartphone or tablet with Endress+Hauser SmartBlue app or PC with DeviceCare from version 1.07.07 or Field Xpert SMT70/SMT77

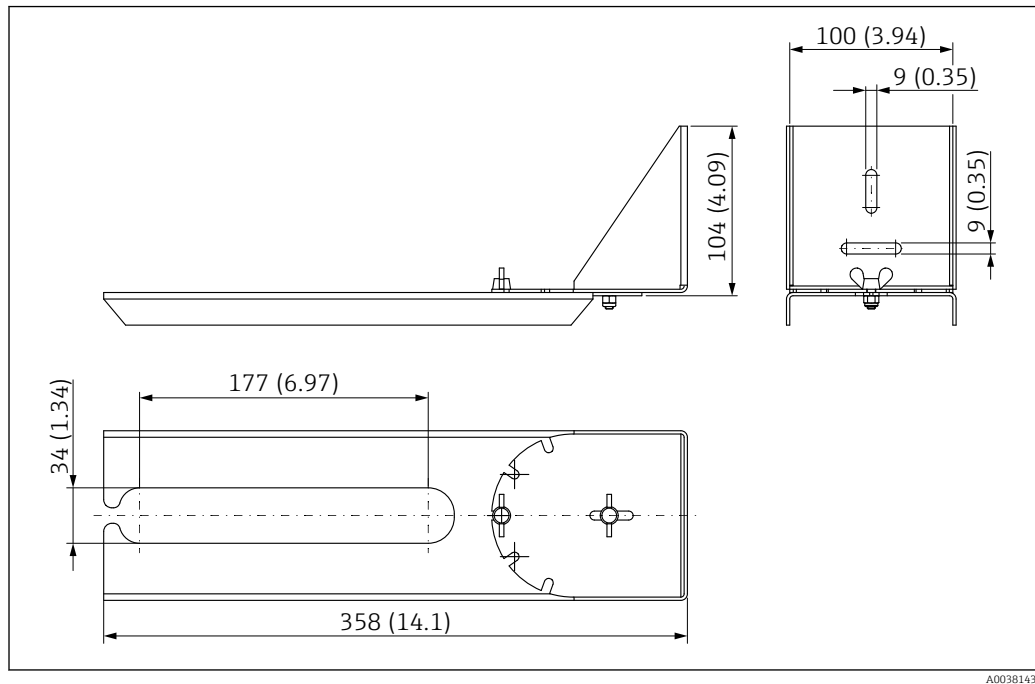
The connection has a range of up to 25 m (82 ft). The range can vary depending on environmental conditions such as attachments, walls or ceilings.

**Supported operating tools**

Smartphone or tablet with Endress+Hauser SmartBlue app, DeviceCare from version 1.07.07, FieldCare, AMS and PDM

**Pivotable mounting bracket**

The pivotable mounting holder is used, for example, to install the device in a manhole over a sewer channel.



51 Dimensions of pivotable mounting bracket. Unit of measurement mm (in)



34 mm (1.34 in) openings for all G 1" or MNPT 1" cable entry thread process connections

**Material**

316L (1.4404)

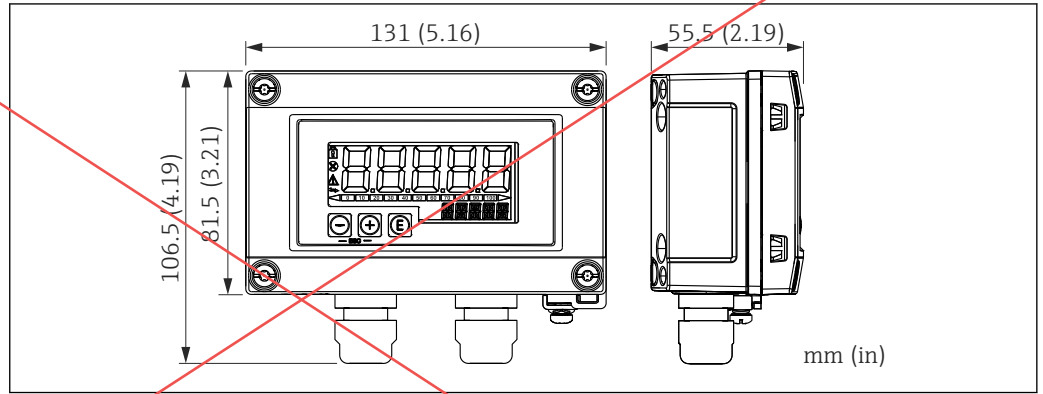
**Order code**

71429910

**FAU40 alignment unit**

The alignment unit is used to optimally align the sensor with the bulk solids.

## RIA15 in the field housing



A0017722

60 Dimensions of RIA15 in the field housing. Unit of measurement mm (in)

**i** Older device versions before February 2025 can only be connected via the 4 to 20 mA current output.

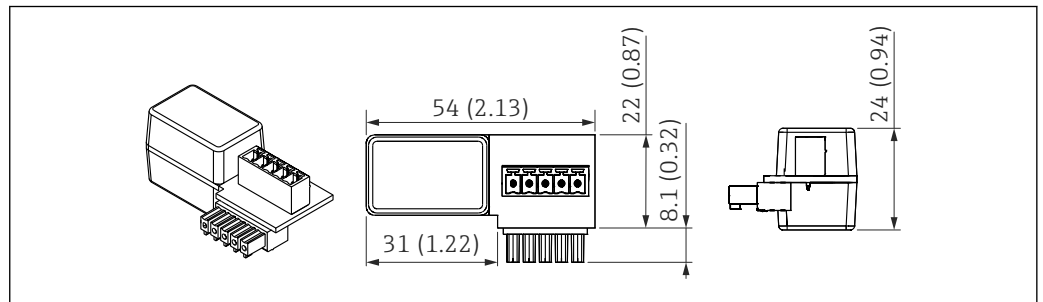
**i** The remote display RIA15 can be ordered with or without operation via the product structure "Accessory enclosed".

Field housing material: Plastic (PBT with steel fibers, antistatic)

Other housing versions are available via the RIA15 product structure.

**i** Alternatively available as an accessory, for details see Technical Information TI01043K and Operating Instructions BA01170K

## HART communication resistor

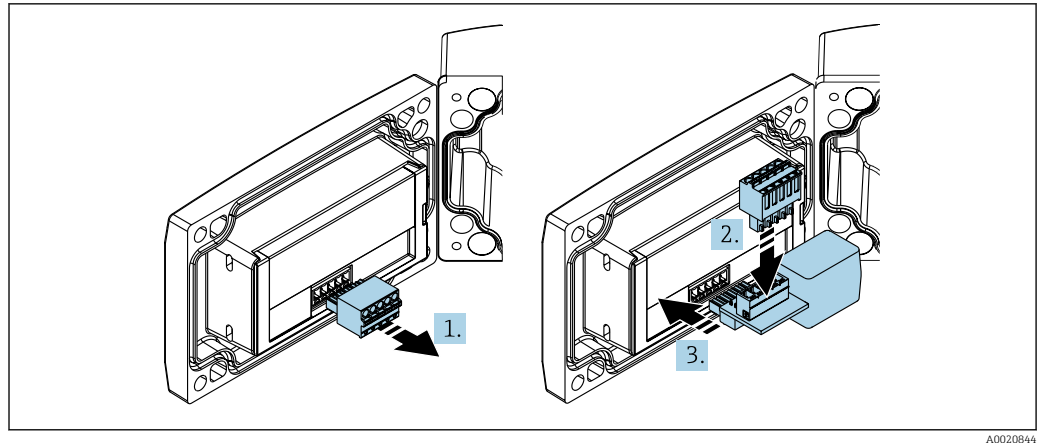


A0020858

61 Dimensions of HART communication resistor. Unit of measurement mm (in)

The HART communication resistor is required for operation of the RIA15 and is supplied with "remote display RIA15, with operation via HART" when ordering.

**i** Document Technical Information TI01043K and Operating Instructions BA01170K



A0020844

**i** In order to operate the RIA15, the HART communication resistor must be integrated.

1. Disconnect the plug-in terminal block.
2. Insert the terminal block into the slot provided on the HART communication resistor module.
3. Insert the HART communication resistor in the slot in the housing.

**DeviceCare SFE100**

Configuration tool for IO-Link, HART, PROFIBUS and FOUNDATION Fieldbusfield devices  
DeviceCare is available for download free of charge at [www.software-products.endress.com](http://www.software-products.endress.com). You need to register in the Endress+Hauser software portal to download the application.

 Technical Information TI01134S

**FieldCare SFE500**

FDT-based plant asset management tool  
It can configure all intelligent field units in your system and helps you manage them. By using the status information, it is also a simple but effective way of checking their status and condition.

 Technical Information TI00028S

**Device Viewer**

All the spare parts for the device, along with the order code, are listed in the *Device Viewer* ([www.endress.com/deviceviewer](http://www.endress.com/deviceviewer)).

**Commubox FXA195 HART**

For intrinsically safe HART communication with FieldCare via the USB interface

 Technical Information TI00404F

**RN22**

Single or two-channel active barrier for safe electrical isolation of 4 to 20 mA standard signal circuits, HART transparent

 Technical Information TI01515K and Operating Instructions BA02004K

**RN42**

Single-channel active barrier with wide-range power supply for safe electrical isolation of 4 to 20 mA standard signal circuits, HART transparent.

 Technical Information TI01584K and Operating Instructions BA02090K

**Field Xpert SMT70**

Universal, high-performance tablet PC for device configuration in Ex Zone 2 and non-Ex areas

 For details, see "Technical Information" TI01342S

**Field Xpert SMT77**

Universal, high-performance tablet PC for device configuration in Ex Zone 1 areas

 For details, see "Technical Information" TI01418S

|                      |                                                                                                                                                                                                                                                           |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SmartBlue app</b> | Mobile app for easy configuration of devices on site via Bluetooth® wireless technology.                                                                                                                                                                  |
| <b>RMA42</b>         | Digital process transmitter for monitoring and displaying analog measured values<br> For details, see Technical Information TI00150R and Operating Instructions BA00287R |

## Documentation



For an overview of the scope of the associated Technical Documentation, refer to the following:

- *Device Viewer* ([www.endress.com/deviceviewer](http://www.endress.com/deviceviewer)): Enter the serial number from the nameplate
- *Endress+Hauser Operations app*: Enter serial number from nameplate or scan matrix code on nameplate.

|                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Standard documentation</b>                       | <p><b>Document type: Operating Instructions (BA)</b><br/>           Installation and initial commissioning – contains all the functions in the operating menu that are needed for a routine measuring task. Functions beyond this scope are not included.</p> <p><b>Document type: Description of Device Parameters (GP)</b><br/>           The document is part of the Operating Instructions and serves as a reference for parameters, providing a detailed explanation of each individual parameter of the operating menu.</p> <p><b>Document type: Brief Operating Instructions (KA)</b><br/>           Quick guide to the first measured value – includes all essential information from incoming acceptance to electrical connection.</p> <p><b>Document type: Safety Instructions, certificates</b><br/>           Depending on the approval, safety instructions are supplied with the device, e.g. XA. This documentation is an integral part of the Operating Instructions.<br/>           Information on the Safety Instructions (XA) that are relevant for the device is provided on the nameplate.</p> |
| <b>Supplementary device-dependent documentation</b> | Additional documents are supplied depending on the device version ordered: Always comply strictly with the instructions in the supplementary documentation. The supplementary documentation is an integral part of the device documentation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Registered trademarks

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### Bluetooth®

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### HART®

Registered trademark of the FieldComm Group, Austin, Texas USA

**Data Sheet 2**

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer:  | Johnston Construction                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Reference: | 32384 Charles County Pomonkey Pump Station Improvements                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Date:      | 06/05/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <u>Qty</u> | <u>Description</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 4          | <p>Mfg: Anchor Scientific: Float Switch, NO<br/>Model Number: GSE60NONC-GOLD</p> <p>Tags/Service:<br/>LSLL-01, LSL-01, LSH-01, LSHH-01 / Pump Station Level Float Switches</p> <p>Specifications:</p> <ul style="list-style-type: none"><li>• G - Model G float switch</li><li>• SE - External weight mounting style</li><li>• 60 - 60ft cable length</li><li>• NONC - 1 normally open, 1 normally closed contact</li><li>• GOLD - For use in intrinsically safe applications</li></ul> |
| 1          | <p>Mfg: SJE Rhombus: Float Switch Mounting Bracket<br/>Model Number: 1009433</p> <p>Tags/Service:<br/>LSLL-01, LSL-01, LSH-01, LSHH-01 / Pump Station Level Float Switches</p> <p>Specifications:</p> <ul style="list-style-type: none"><li>• 4 float bracket</li><li>• Supplied with cord strain relief connectors</li><li>• 300 series stainless steel construction</li><li>• Strain relief connector accomodates 18/2, 18/3, 16/2, 16/3 SJOW and SJTW cords.</li></ul>               |

# Float Level Switches



Listed



anchor scientific inc.



Priced From \$47.00



Priced From \$47.00

## roto-float

TYPE S - SUSPENDED &amp; TYPE P - PIPE MOUNTED

### TYPES - SUSPENDED & TYPE P - PIPE MOUNTED

The ROTO-FLOAT is a direct acting float switch. Each ROTO-FLOAT contains a single pole mercury switch which actuates when the longitudinal axis of the float is horizontal, and deactuates when the liquid level falls 1" below the actuation elevation.

The float is a chemical resistant polypropylene casing with a firmly bonded electrical cable protruding. One end of the cable is permanently connected to the enclosed mercury switch and the entire assembly is encapsulated to form a completely watertight and impact resistant unit. Type S - Suspended has built-in weight.

ROTO-FLOATS can be mounted on a support pipe (type P) or suspended from above (type S). Advantages of the ROTO-FLOAT are low cost, simplicity and reliability.

NOTE: Mercury switches are not to be used in potable water.

### MATERIALS OF CONSTRUCTION

Float housing.....Polypropylene

Cable clamp.....Polypropylene

Cable jacket.....P.V.C

### Applications

- Pilot Duty
- Industrial Control Equipment

### CABLE

P.V.C. type STO #18 conductors (41 strand) rated 600 volts

- Various lengths available
- See table of models
- Non-standard lengths also available on special order.

### Ordering information

| Switch Arrangement | Cable Length | Suspended Type S Model No. | Pipe Mounted Type P Model |
|--------------------|--------------|----------------------------|---------------------------|
| No.                |              |                            |                           |
| Normally           | 20           | S20NO                      | P20NO                     |
| Open               | 30           | S30NO                      | P30NO                     |
|                    | 40           | S40NO                      | P40NO                     |
| Normally           | 20           | S20NC                      | P20NC                     |
|                    | 30           | S30NC                      | P30NC                     |

## eco-float

Model G

### Applications

The Eco-Float can be used in a variety of liquid level monitoring applications including sumps, sewage ejectors, septic tanks, vaults, lift stations, and tanks. Eco-Floats are ruggedly constructed of corrosion resistant materials, enabling them to be used in a variety of different liquids. Some applications are subject to additional requirements described in the National Electric Code.

### Description

The Eco-float is a **mercury-free** float switch for controlling liquid levels in a variety of applications. A snap-action switch is activated by a steel ball rolling back and forth within a switching tube in a plastic float housing. There is a minimum differential between "on" and "off" of approximately 3.5 inches. Greater differentials can be achieved when the pipe mounted or externally

### FEATURES

- Mercury Free
- Variety of Mounting Styles
- Variety of Circuit Configurations
- Installation Easy
- Differential In One Float
- Replaces Diaphragm and Mercury Switches

### Ordering information

| Model | Mounting Style                                                   | Cable Length*                          | Circuit Configuration                                                              |
|-------|------------------------------------------------------------------|----------------------------------------|------------------------------------------------------------------------------------|
| G     | SE (external weight)<br>SI (internal weight)<br>P (pipe-mounted) | 10"<br>15"<br>20"<br>30"<br>40"<br>50" | NO (normally open)<br>NC (normally closed)<br>NONC (normally open/normally closed) |

\* Custom lengths available.

Example: GSI 20NO - Eco-Float, suspended internal wt. 20', normally opens contacts.

60 - 60ft cable length

## solo-float

### DESCRIPTION

The new Solo-Float" is a direct acting **mercury-free** float switch for controlling 1 1/2 HP and smaller pump motors. The new Solo-Float has been completely redesigned and uses a metal ball rolling back and forth in a tube to actuate a reliable, snap-action switch. An adjustable, plastic clamp with stainless steel hose clamp is provided for attaching the Solo-Float at the appropriate level on the pump discharge pipe. By varying the tether length, a pumping range of 8" - 14" can be achieved. The Solo-Float is available with a piggyback plug in either a 115 volt or 230 volt configuration. It is also available blunt cut, without a plug, for wiring to an Anchor Scientific JX junction box.

### FEATURES

- Reliability
- Adjustable Pumping Range
- Manual Operation
- Non-mercury Element
- Converts Nonautomatic Pumps To Automatic
- Replaces Diaphragm Pressure Switches
- Easy Installation
- In Stock, Low Cost

Example:

| Model | Cable Length* | Circuit Configuration | Plug Voltage     |
|-------|---------------|-----------------------|------------------|
| D     | 20'           | NO (normally open)    | 115 for 115 volt |

### Ordering information

| Model | Cable Length*                   | Circuit Configuration                      | Plug Voltage                         |
|-------|---------------------------------|--------------------------------------------|--------------------------------------|
| D     | 10'<br>15'<br>20'<br>25'<br>30' | NO (normally open)<br>NC (normally closed) | 115 for 115 volt<br>230 for 230 volt |

\* Custom lengths available.

# FLOAT ACCESSORIES

To Finish Any Onsite Installation

## FLOAT SWITCH MOUNTING BRACKET

TAGS: LSLL-01, LSL-01, LSH-01, LSHH-01

### CABLE CLAMP ASSEMBLY

Provides cable anchor point and strain relief. Secures cable in vertical direction.

- For use with 18/16 gauge, 14 gauge, & 12 gauge cable

**MOUNTING CLAMP:** ABS Nylon blend

**HOSE CLAMP:** highly corrosion resistant stainless steel



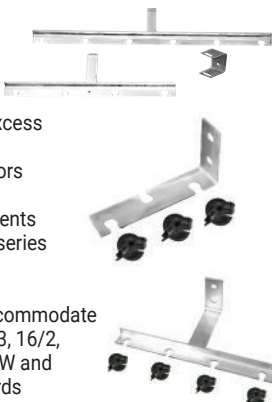
### FLOAT BRACKET

Provides convenient installation of float switches.

- Six Float Bracket supports 1-6 floats
- Four Float Bracket supports 1-4 floats
- Can be directly mounted to wall or used with wall mounting device
- Provides convenient location to stow excess float cord
- Supplied with cord strain relief connectors

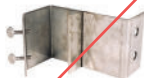
**METAL COMPONENTS:** all metal components are made of 300 series stainless steel

**STRAIN RELIEF CONNECTORS:** which accommodate 18/2, 18/3, 16/2, 16/3 SJOW and SJTW cords



### PIPE MOUNTING BRACKET

- Secures discharge pipe to basin wall.
- Clamps directly to 2" PVC discharge pipe.



### CABLE WEIGHT

Provides an accurate pivot point for suspended float switches.

- Gripper teeth on clip and weight channel securely lock float cable into place
- Cable weight can be adjusted without the use of tools

**HOUSING:** 1 pound, 5 ounce (0.6 kgs.), 2.8 inch diameter x 3.3 inch long (7.1 cm diameter x 8.4 cm), impact resistant, non-corrosive, PVC housing for use in liquids up to 140°F (60°C)

**CLIP:** injection molded acetal plastic

**WIRE/CABLE ACCOMMODATED:** SJOW, SJTW, 18/2, 18/3, 16/2, 16/3, 14/2 and 14/3

**SHIP WEIGHT:** 1.6 lbs. (.726 kgs)



U.S. Patent  
5,306,885

### CABLE WEIGHT MINI

Provides an accurate pivot point for suspended float switches.

- One piece cable weight
- Turn and Lock for easy installation

**HOUSING:** 9.6 ounces (0.27 kgs.), 2.5 inch diameter x 2.5 inch long (6.35 cm diameter x 6.35 cm), impact resistant, non-corrosive, PVC housing for use in liquids up to 140°F (60°C)

**WIRE/CABLE ACCOMMODATED:** SJOW, SJTW, 18/2, 18/3

**NOTE:** The Cable Weight Mini can be used on the following control switches with 18/2 or 18/3 cable:

- SJE SignalMaster®
- SJE SignalMaster® SPDT
- SJE MilliAmpMaster™
- Sensor Float® Mini



### ELAPSED TIME METER WITH PIGGY-BACK PLUG

Plugs into switched outlet or piggy-back switch of plugger control panel. Unit must be placed inside an enclosure.

- Screws and mounting adhesive included



### VERTICAL REED SWITCH

- Control differential: .375 inch (1 cm)
- Maximum angle from vertical: 5°
- Electrical: 250 mAmps, 12-125 VAC, 50/60 Hz  
500 mAmps, 6-12 VDC, 50/60 Hz
- Housing and Float: 1.60 inch diameter x 6.7 inch long (4.06 cm x 17.01 cm), high impact, corrosion resistant PVC for use in sewage and non-potable water up to 120°F (50°C)



### ANCHOR

- 10 lb. weight
- Yellow, vinyl coated, mushroom style



### CHAIN

- 1/8" thickness
- 316 grade stainless steel
- Available in one foot increments



### 6 HOOK BRACKET

- Universal hanger for up to 6 float switches, submersible transmitters, and small horsepower pump cords
- Hook diameter: 0.188" x 3.0" (1.8 mm x 76.2 mm)
- 304 stainless steel



**FLOAT ACCESSORIES** - Cable Clamp Assembly, Float Brackets, Pipe Mounting Bracket, Cable Weights, ETM w/ Piggy-Back Plug, Vertical Reed Switch, Anchor and Chain.

| Part #             | Description                                                             |
|--------------------|-------------------------------------------------------------------------|
| <del>1008569</del> | <del>18 &amp; 16 Gauge Cable Clamp Assembly</del>                       |
| <del>1008574</del> | <del>14 Gauge Cable Clamp Assembly</del>                                |
| <del>1008575</del> | <del>12 Gauge Cable Clamp Assembly</del>                                |
| <del>1027830</del> | <del>304 Stainless Steel Float Bracket with 3 Black Cord Snubbers</del> |
| <del>1027647</del> | <del>304 Stainless Steel Float Bracket with 4 Black Cord Snubbers</del> |
| <del>1009432</del> | <del>4 Float Bracket with Mounting Device</del>                         |
| 1009433            | 4 Float Bracket without Mounting Device ←                               |
| <del>1009434</del> | <del>6 Float Bracket with Mounting Device</del>                         |
| <del>1009435</del> | <del>6 Float Bracket without Mounting Device</del>                      |
| <del>1009436</del> | <del>Float Bracket Mounting Device</del>                                |
| <del>1109826</del> | <del>Pipe Mounting Bracket</del>                                        |
| <del>1002230</del> | <del>Cable Weight</del>                                                 |
| <del>1030356</del> | <del>Cable Weight Mini</del>                                            |
| <del>1022846</del> | <del>Elapsed Time Meter w/120V Piggy-Back Plug</del>                    |
| <del>1032834</del> | <del>Elapsed Time Meter w/230V Piggy-Back Plug</del>                    |
| <del>1011027</del> | <del>10 foot Vertical Reed Switch 120V Box</del>                        |
| <del>1016561</del> | <del>15 foot Vertical Reed Switch 120V Box</del>                        |
| <del>1018687</del> | <del>10 lb. Mushroom Anchor</del>                                       |
| <del>1018823</del> | <del>1/8 inch Chain</del>                                               |
| <del>1052814</del> | <del>6 Hook Bracket</del>                                               |

**Data Sheet 4**

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer:  | Johnston Construction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Reference: | 32384 Charles County Pomonkey Pump Station Improvements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Date:      | 06/05/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <u>Qty</u> | <u>Description</u>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1          | <p>Mfg: E+H: Digital Process Meter Display<br/>Model Number: RIA452-C211A11A</p> <p>Tags/Service:<br/>LEVEL DISPLAY / PLC/RTU Panel Process Meter Display</p> <p>Specifications:</p> <ul style="list-style-type: none"><li>• RIA452 - Process indicator with pump control unit</li><li>• C - FM AIS, I, II, III/2/ABCDEFGH approvals</li><li>• 2 - 20-30VDC, 20-28VAC power supply</li><li>• 1 - 0/4-20mA measuring signal</li><li>• 1 - 4 output relays, SPDT</li><li>• A - Standard communication</li><li>• 1 - 96x96 panel mounting housing, front IP65</li><li>• 1 - Basic version</li><li>• A - Standard version</li></ul> |

## DIGITAL PROCESS METER DISPLAY

### TAGS: LEVEL DISPLAY

# Technical Information

## RIA452

### Process indicator



Digital process indicator in panel mounted housing for monitoring and displaying analog measured values with pump control and batch functions and flow calculation

#### Application

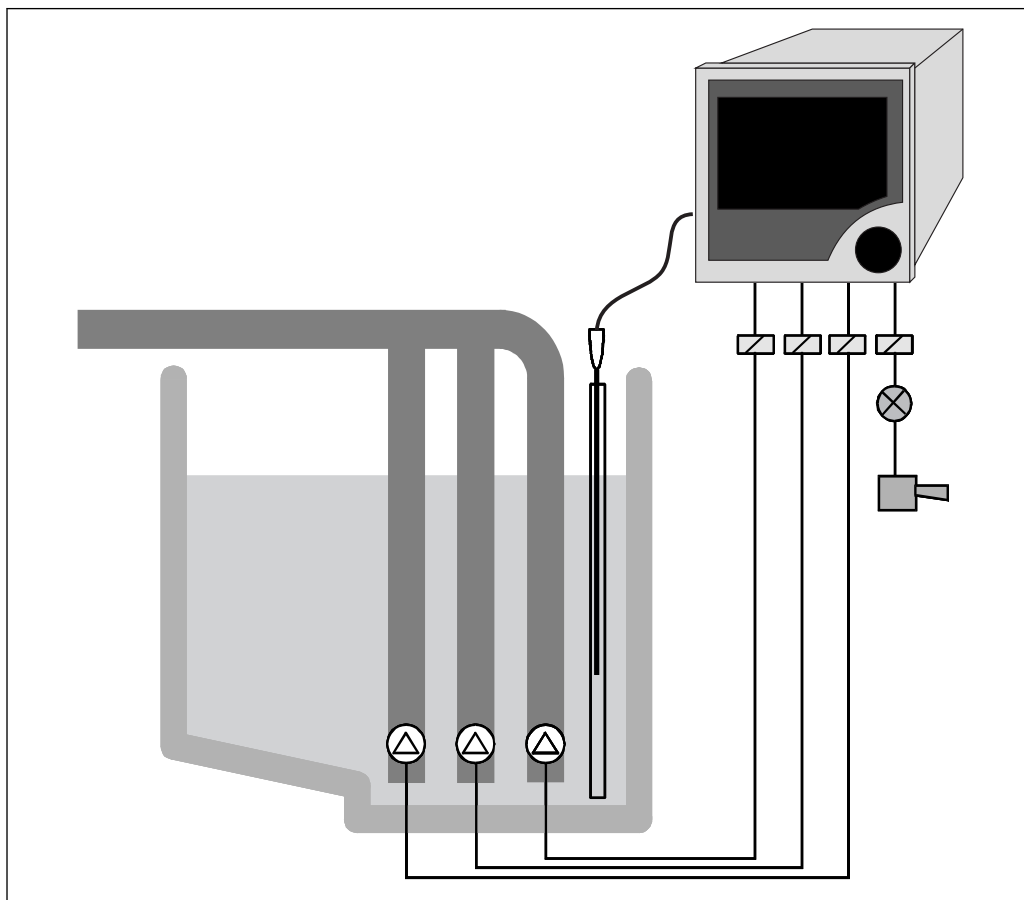
- Water/wastewater sector
- ~~Power industry~~
- ~~Raw materials~~
- ~~Chemicals industry~~
- ~~Food industry~~

#### Your benefits

- 7-digit 14-segment LC display
- Multicolored
- Large bar graph with overrange and underrange
- Intrinsically safe input with transmitter power supply
- Digital status inputs for pump monitoring
- Universal input
- Up to eight relays
- Min/Max value saved
- Pump control functions
- Batch functions
- Flow measurement for open channels and weirs
- Linearization table with 32 support points
- Analog output
- Pulse output with totalizer
- Jog-shuttle operation
- Freely programmable units
- Configuration via interface and operating software
- Tank linearization via PC software

## Function and system design

### Measuring principle



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1 Example of an application of the process indicator

The single-channel RIA452 process indicator monitors and displays analog measured values. Pumps can be monitored with the digital status inputs. The measured value is displayed using the seven-digit, 14-segment LC display. Numbers and units are displayed in white, the bar graph in yellow, overrange and underrange in red, and the limit value flags and digital status inputs in green and yellow. The RIA452 can provide power directly to two-wire transmitters that are connected. As an option, it is also possible to select the input and the transmitter power supply as intrinsically safe for Ex applications. Up to eight freely programmable relays monitor the measured value for limit value overshoot and undershoot. Other operating modes for the relays include sensor or device malfunction, batch and pump control functions (e.g. alternating pump control). Furthermore, the RIA452 can be used as a preset counter and for measuring flow at open channels and measuring weirs.

The scalable analog output offers many different ways of forwarding the input signal: zoom function, linearization, offset, inversion and signal conversion (input/output conversion). The optional pulse output allows users to output integrated process values.

### Measuring system

Microcontroller-controlled indicator in a panel-mounted housing with a multicolored, illuminated LC display. Analog measured value acquisition is via an analog/digital converter. The digital status inputs are scanned cyclically. Power can be supplied directly to two-wire sensors with the transmitter power supply which is integrated as standard. The current input is optionally available as an intrinsically safe version for Ex applications. In this case, the RIA452 has a second, intrinsically safe transmitter power supply.

The freely scalable analog output is output via digital/analog conversion. The digital pulse output is output directly.

Up to eight relays are available in the device for monitoring limit values, pump control and batch functions.

The device can be operated either onsite via the jog/shuttle dial, or via the PC with an operating software. Operation can be locked using the hardware key or software code.

### Linearization

The following flow curves are programmed into the device for open channels and weirs:

- Khafagi-Venturi flume
- ISO Venturi flume
- BST <sup>1)</sup> Venturi flume
- Parshall flume
- Palmer-Bowlus flume
- Rectangular weir
- Rectangular weir with constriction
- NFX <sup>2)</sup> rectangular weir
- NFX <sup>2)</sup> rectangular weir with constriction
- Trapezoidal weir
- Triangular weir
- BST <sup>1)</sup> triangular weir
- NFX <sup>2)</sup> triangular weir

User-configurable flow formula

$$Q = C * (h^{\alpha} + \gamma * h^{\beta})$$

The parameters  $\alpha$ ,  $\beta$ ,  $\gamma$  and C can be entered freely.

### Linearization function

Up to 32 user-definable linearization points are available in the device for the linearization of the input, e.g. for tank linearization.

The linearization table for standard tanks and customer-specific tanks can be generated with the ReadWin 2000 operating software.

## Input

### Measured variable

- Current (standard)
- Digital inputs (standard)
- Current/voltage, resistance, RTD assembly, thermocouples (universal input option)

### Measuring range

#### Current input:

Current:

- 0/4 to 20 mA +10% overrange, 0 to 5 mA
- Short-circuit current: max. 150 mA
- Input impedance:  $\leq 5 \Omega$
- Response time:  $\leq 100 \text{ ms}$

#### Universal input:

Current:

- 0/4 to 20 mA + 10% overrange, 0 to 5 mA
- Short-circuit current: max. 100 mA
- Input impedance:  $\leq 50 \Omega$

Voltage:

- $\pm 150 \text{ mV}$ ,  $\pm 1 \text{ V}$ ,  $\pm 10 \text{ V}$ ,  $\pm 30 \text{ V}$ , 0 to 100 mV, 0 to 200 mV, 0 to 1 V, 0 to 10 V
- Input impedance:  $\geq 100 \text{ k}\Omega$

Resistance:

30 to 3000  $\Omega$  in 3/4-wire technology

RTD assembly:

- Pt100/500/1000, Cu50/100, Pt50 in 3/4-wire technology
- Measuring current for Pt100/500/1000 = 0.25 mA

1) BST: British Standard

2) NFX: French standard NFX 10-311

~~Thermocouple types:~~

- J, K, T, N, B, S, R as per IEC584
- D, C as per ASTM E998
- U, L as per DIN43710/GOST
- Response time:  $\leq 100$  ms

**Digital input:**

## Digital input:

- Voltage level -3 to 5 V low, 12 to 30 V high (as per DIN19240)
- Input voltage max. 34.5 V
- Input current typ. 3 mA with overload and reverse polarity protection
- Sampling frequency max. 10 Hz

**Galvanic isolation**

Towards all other circuits

## Output

**Output signal**

- Relay, transmitter power supply (standard)
- Current, voltage, pulse, intrinsically safe transmitter power supply (option)

**Signal on alarm**

No measured value visible on the LC display, no background illumination, no sensor power supply, no output signals, relays behave in safety-oriented manner.

**Current/voltage output**

Analog output range:  
0/4 to 20 mA (active), 0 to 10 V (active)

## Load:

- $\leq 600 \Omega$  (current output)
- Max. output current 22 mA (voltage output)

## Signal characteristics:

Freely scalable signal

Galvanic isolation towards all other circuits

**Pulse output (open collector)**

## Pulse output (open collector):

- Frequency range to 2 kHz
- $I_{\max} = 200$  mA
- $U_{\max} = 28$  V
- $U_{\text{low/max}} = 2$  V at 200 mA
- Pulse width = 0.04 to 2 000 ms

**Relay output**

## Signal characteristics:

Binary, switches when the limit value is reached

Switch function: limit relay switches for the operating modes:

- Minimum/maximum safety
- Alternating pump control function
- Batch function
- Time control
- Window function
- Gradient
- Device malfunction
- Sensor malfunction

## Switching threshold:

Freely programmable

## Hysteresis:

0 to 99%

## Signal source:

- Analog input signal
- Integrated value
- Digital input

Number:

4 in basic unit (~~can be extended to 8 relays, option~~)

Electrical specifications:

- Relay type: changeover
- Relay switching capacity: 250 V<sub>AC</sub> / 30 V<sub>DC</sub>, 3 A
- Switch cycles: typically 10<sup>5</sup>
- Switching frequency: max. 5 Hz
- Minimum switching load: 10 mA / 5 V<sub>DC</sub>

Galvanic isolation towards all other circuits



Mixed assignment of low and extra-low voltage circuits is not permitted for neighboring relays.

---

#### Transmitter power supply

##### **Transmitter power supply 1, terminal 81/82 (optionally intrinsically safe):**

Electrical specifications:

- Output voltage: 24 V ±15%
- Output current: max. 22 mA (for U<sub>out</sub> ≥ 16 V, sustained short-circuit proof)
- Impedance: ≤ 345 Ω

##### **Transmitter power supply 2, terminal 91/92:**

Electrical specifications:

- Output voltage: 24 V ±15%
- Output current: max. 250 mA (sustained short-circuit proof)

##### **Transmitter power supply 1 and 2:**

Galvanic isolation:

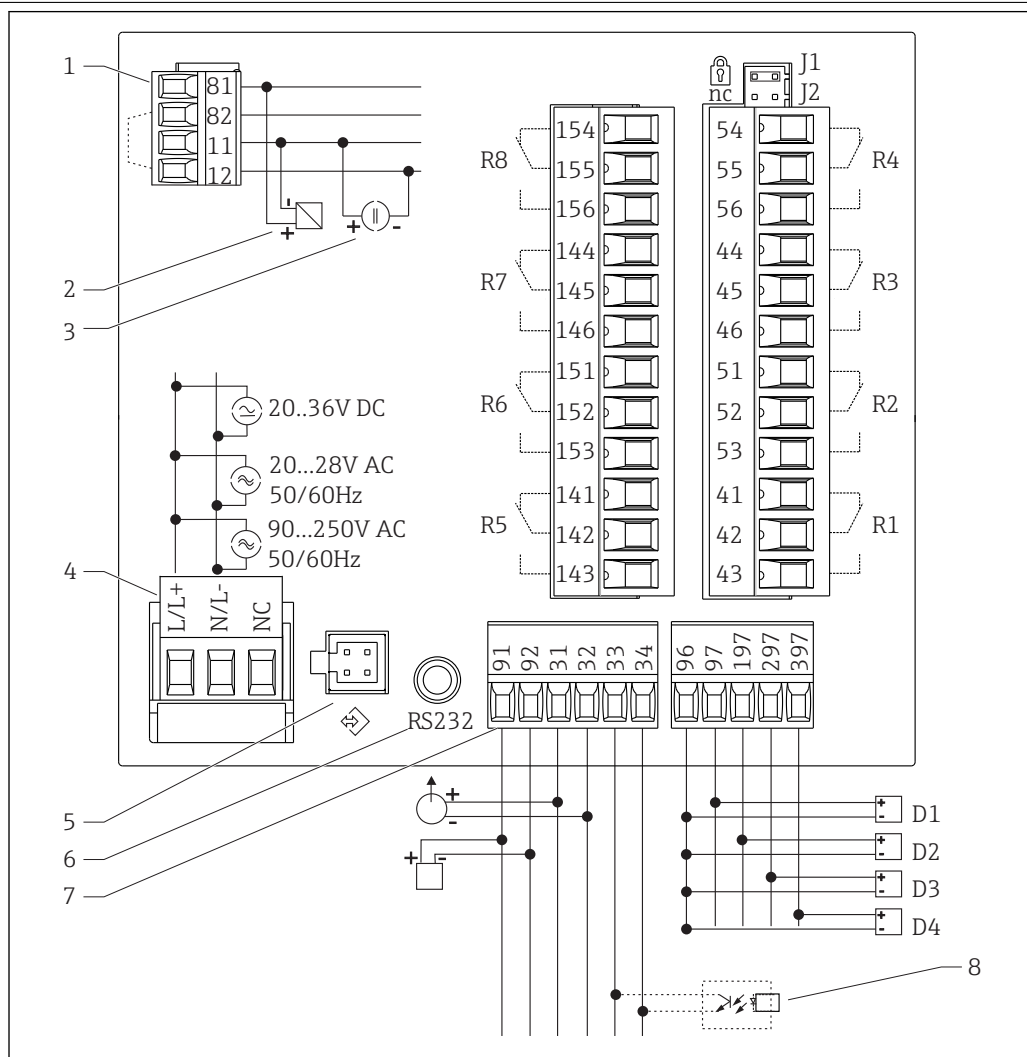
Towards all other circuits

##### **HART®**

HART® signals are not affected

## Power supply

### Terminal assignment

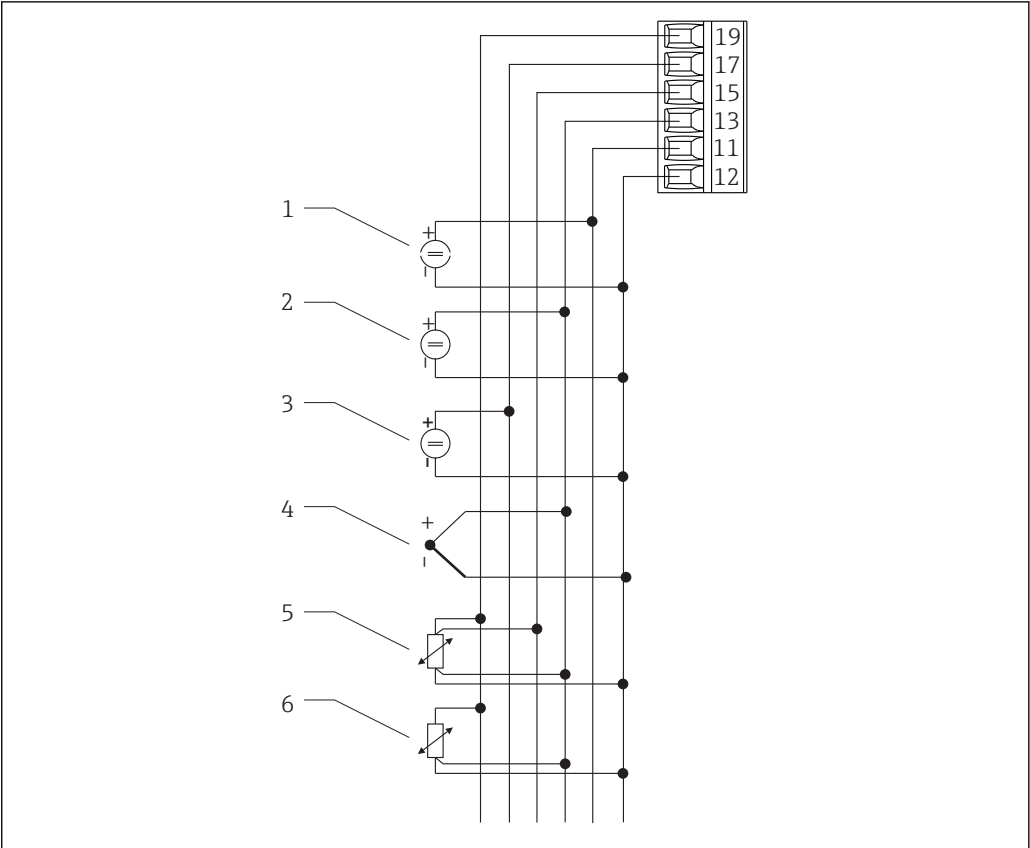


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2 Terminal assignment of process indicator

- |   |                                               |    |                                            |
|---|-----------------------------------------------|----|--------------------------------------------|
| 1 | Current input (12 and 82 jumpered internally) | 7  | Transmitter power supply and analog output |
| 2 | - passive sensor                              | 8  | Open collector output                      |
| 3 | - active sensor                               |    | D1 to D4 Digital inputs                    |
| 4 | Power supply                                  |    | R1 to R4 Relay outputs                     |
| 5 | Interface for PC operating software           |    | R5 to R8 Relay outputs (optional)          |
| 6 | RS232 interface                               | J1 | Hardware write protection                  |

Universal input option



3 Universal input terminal assignment

- |                              |                        |
|------------------------------|------------------------|
| 1 Current input 0/4 to 20 mA | 4 Thermocouples        |
| 2 Voltage input ±1 V         | 5 RTD assembly, 4-wire |
| 3 Voltage input ±30 V        | 6 RTD assembly, 3-wire |

Interface connection data

RS232

- Connection: jack socket 3.5 mm, rear of device
- Transmission protocol: ReadWin 2000
- Transmission rate: 38 400 Baud

|                |                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supply voltage | <ul style="list-style-type: none"><li>■ Low voltage power unit 90 to 250 V<sub>AC</sub> 50/60 Hz</li><li>■ Extra-low voltage power unit 20 to 36 V<sub>DC</sub> or 20 to 28 V<sub>AC</sub> 50/60 Hz</li></ul> <p>The device must be powered only by a power unit that operates using a limited energy circuit in accordance with UL/EN/IEC 61010-1, Section 9.4 and the requirements in Table 18.</p> |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                   |                              |
|-------------------|------------------------------|
| Power consumption | Power consumption max. 24 VA |
|-------------------|------------------------------|

Performance characteristics

|                                |                                                                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Reference operating conditions | <p>Power supply: 230 V<sub>AC</sub> ±10%, 50 Hz ±0.5 Hz</p> <p>Warm-up period: 90 min</p> <p>Ambient temperature: 25 °C (77 °F)</p> |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

**Maximum measured error****Current input**

|                   |                     |
|-------------------|---------------------|
| Accuracy          | 0.1% of full scale  |
| Resolution        | 13 bit              |
| Temperature drift | ≤ 0.4%/10 K (18 °F) |

**Universal input**

|                                                                  | Input:                 | Range:                                                                                                                                                                                        | Maximum measured error of measuring range (oMR):                                   |
|------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Accuracy                                                         | Current                | 0 to 20 mA, 0 to 5 mA, 4 to 20 mA; overrange: to 22 mA                                                                                                                                        | ±0.10%                                                                             |
|                                                                  | Voltage > 1 V          | 0 to 10 V, ±10 V, ±30 V                                                                                                                                                                       | ±0.10%                                                                             |
|                                                                  | Voltage ≤ 1 V          | ±1 V, 0 to 1 V, 0 to 200 mV, 0 to 100 mV, ±150 mV                                                                                                                                             | ±0.10%                                                                             |
|                                                                  | Resistance thermometer | Pt100, −200 to 600 °C (−328 to 1 112 °F) (IEC751, JIS1604, GOST)<br>Pt500, −200 to 600 °C (−328 to 1 112 °F) (IEC751, JIS1604)<br>Pt1000, −200 to 600 °C (−328 to 1 112 °F) (IEC751, JIS1604) | 4-wire: ± (0.10% oMR + 0.3 K (0.54 °F))<br>3-wire: ± (0.15% oMR + 0.8 K (1.44 °F)) |
|                                                                  |                        | Cu100, −200 to 200 °C (−328 to 392 °F) (GOST)<br>Cu50, −200 to 200 °C (−328 to 392 °F) (GOST)<br>Pt50, −200 to 600 °C (−328 to 1 112 °F) (GOST)                                               | 4-wire: ± (0.20% oMR + 0.3 K (0.54 °F))<br>3-wire: ± (0.20% oMR + 0.8 K (1.44 °F)) |
|                                                                  |                        | Resistance measurement                                                                                                                                                                        | 30 to 3 000 Ω                                                                      |
|                                                                  | Thermocouples          | Typ J (Fe-CuNi),<br>−210 to 999.9 °C (−346 to 1 382 °F) (IEC584)                                                                                                                              | ± (0.15% oMR + 0.5 K (0.9 °F)) from<br>−100 °C (−148 °F)                           |
|                                                                  |                        | Typ K (NiCr-Ni),<br>−200 to 1 372 °C (−328 to 2 502 °F) (IEC584)                                                                                                                              | ± (0.15% oMR + 0.5 K (0.9 °F)) from<br>−130 °C (−234 °F)                           |
|                                                                  |                        | Typ T (Cu-CuNi), −270 to 400 °C (−454 to 752 °F) (IEC584)                                                                                                                                     | ± (0.15% oMR + 0.5 K (0.9 °F)) from<br>−200 °C (−328 °F)                           |
|                                                                  |                        | Typ N (NiCrSi-NiSi),<br>−270 to 1 300 °C (−454 to 2 372 °F) (IEC584)                                                                                                                          | ± (0.15% oMR + 0.5 K (0.9 °F)) from<br>−100 °C (−148 °F)                           |
|                                                                  |                        | Typ B (Pt30Rh-Pt6Rh),<br>0 to 1 820 °C (32 to 3 308 °F) (IEC584)                                                                                                                              | ± (0.15% oMR + 1.5 K (2.7 °F)) from<br>600 °C (1 112 °F)                           |
|                                                                  |                        | Typ D (W3Re/W25Re),<br>0 to 2 315 °C (32 to 4 199 °F) (ASTME998)                                                                                                                              | ± (0.15% oMR + 1.5 K (2.7 °F)) from<br>500 °C (932 °F)                             |
|                                                                  |                        | Typ C (W5Re/W26Re),<br>0 to 2 315 °C (32 to 4 199 °F) (ASTME998)                                                                                                                              | ± (0.15% oMR + 1.5 K (2.7 °F)) from<br>500 °C (932 °F)                             |
|                                                                  |                        | Typ L (Fe-CuNi),<br>−200 to 900 °C (−328 to 1 652 °F) (DIN43710, GOST)                                                                                                                        | ± (0.15% oMR + 0.5 K (0.9 °F)) from<br>−100 °C (−148 °F)                           |
| Typ U (Cu-CuNi),<br>−200 to 600 °C (−328 to 1 112 °F) (DIN43710) |                        | ± (0.15% oMR + 0.5 K (0.9 °F)) from<br>−100 °C (−148 °F)                                                                                                                                      |                                                                                    |
| Typ S (Pt10Rh-Pt), 0 to 1 768 °C (32 to 3 214 °F) (IEC584)       |                        | ± (0.15% oMR + 3.5 K (6.3 °F)) for<br>0 to 100 °C (32 to 212 °F)<br>± (0.15% oMR + 1.5 K (2.7 °F)) for<br>100 to 1 768 °C (212 to 3 214 °F)                                                   |                                                                                    |
| Typ R (Pt13Rh-Pt),<br>−50 to 1 768 °C (−58 to 3 214 °F) (IEC584) |                        | ± (0.15% oMR + 1.5 K (2.7 °F)) for<br>100 to 1 768 °C (212 to 3 214 °F)                                                                                                                       |                                                                                    |
| Resolution                                                       |                        | 16 bit                                                                                                                                                                                        |                                                                                    |
| Temperature drift                                                |                        | Temperature drift: ≤ 0.1%/10 K (18 °F)                                                                                                                                                        |                                                                                    |

**Current output**

|                   |                                                            |
|-------------------|------------------------------------------------------------|
| Linearity         | 0.1% of full scale                                         |
| Resolution        | 13 bit                                                     |
| Temperature drift | Temperature drift: $\leq 0.1\%/10\text{ K}$ (18 °F)        |
| Output Ripple     | 10 mV at 500 $\Omega$ for frequencies $\leq 50\text{ kHz}$ |

**Voltage output**

|                   |                                                     |
|-------------------|-----------------------------------------------------|
| Linearity         | 0.1% of full scale                                  |
| Resolution        | 13 bit                                              |
| Temperature drift | Temperature drift: $\leq 0.1\%/10\text{ K}$ (18 °F) |

## Installation

---

|                          |                                                                           |
|--------------------------|---------------------------------------------------------------------------|
| <b>Mounting location</b> | Panel, cut-out 92 x 92 mm (3.62x3.62 in) (see 'Mechanical construction'). |
| <b>Orientation</b>       | Horizontal +/- 45° in every direction                                     |

---

## Environment

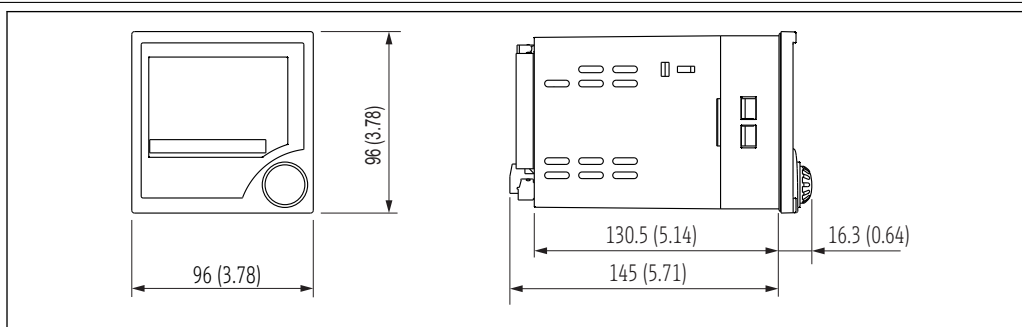
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|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ambient temperature range</b>           | -20 to 60 °C (-4 to 140 °F)                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Storage temperature</b>                 | -30 to 70 °C (-22 to 158 °F)                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Operating altitude</b>                  | < 3 000 m (9 840 ft) over MSL                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Climate class</b>                       | To IEC 60654-1, Class B2                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Degree of protection</b>                | Front IP 65 / NEMA 4<br>Device casing IP 20                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Shock and vibration resistance</b>      | 2 Hz (+3/-0) ... 13.2 Hz: $\pm 1\text{ mm}$ ( $\pm 0.04\text{ in}$ )<br>13.2 to 100 Hz: 0.7 g                                                                                                                                                                                                                                                                                                                                                     |
| <b>Electromagnetic compatibility (EMC)</b> | <b>CE compliance</b><br>Electromagnetic compatibility in accordance with all the relevant requirements of the IEC/EN 61326 series and NAMUR Recommendation EMC (NE21). For details refer to the EU Declaration of Conformity.<br><br>Maximum measured error <1% of measuring range.<br><br>Interference immunity as per IEC/EN 61326 series, industrial requirements.<br><br>Interference emission as per IEC/EN 61326 series, Class A equipment. |
| <b>Electrical protection class</b>         | IEC 60529 (IP code) / NEMA 250                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Condensation</b>                        | Front: permitted<br>Device casing: not permitted                                                                                                                                                                                                                                                                                                                                                                                                  |

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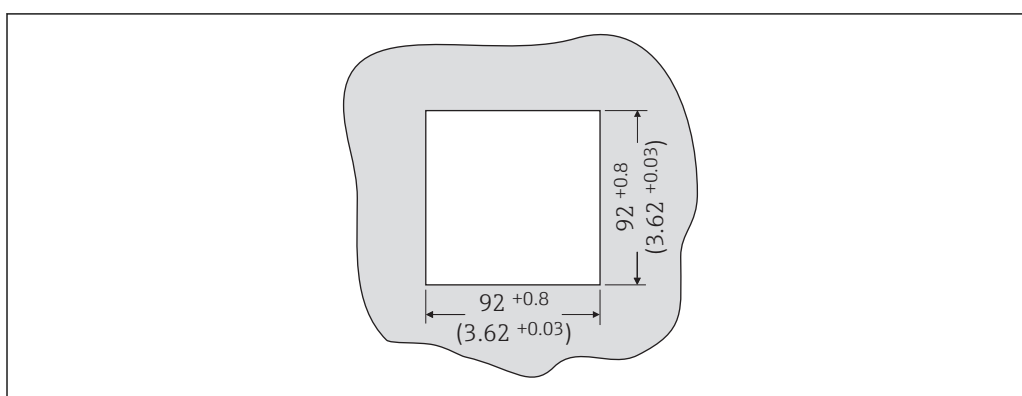
## Mechanical construction

### Design, dimensions



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4 Dimensions of the process indicator in mm (in)



A0028476

5 Panel cutout, dimensions in mm (in)

**Weight** 500 g (17.64 oz)

**Materials**

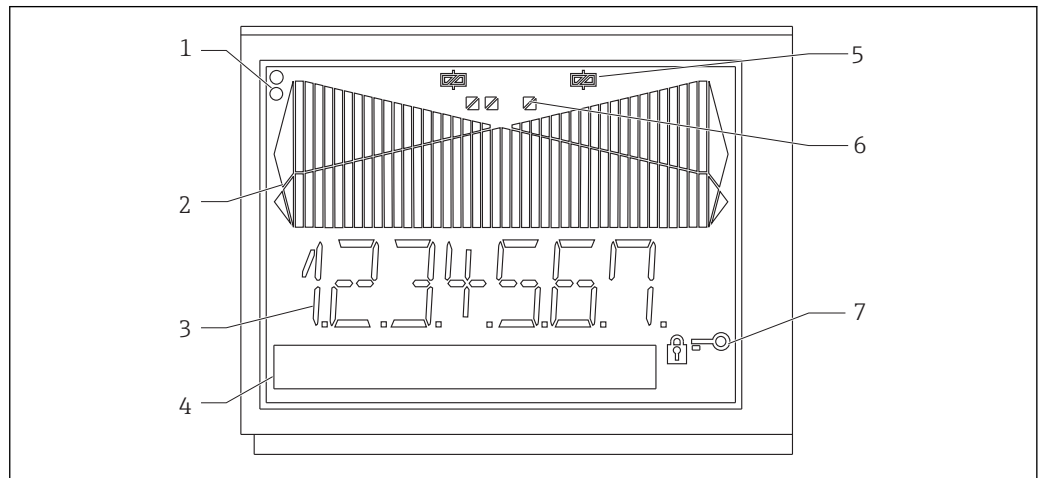
- Housing front: ABS plastic
- Housing casing: ABS GF plastic

**Terminals** Plug-in screw terminals, clamping range 1.5 mm<sup>2</sup> (16 AWG) solid, 1 mm<sup>2</sup> (18 AWG) strand with wire ferrule

## Operability

### Local operation

#### Display elements



A0028477

#### 6 Display elements of the process indicator

- 1 Device status LEDs: green - device ready for operation; red - device or sensor malfunction
- 2 Bar graph with overrange and underrange
- 3 7-digit 14-segment display
- 4 Unit and text field 9x7 dot matrix
- 5 Relay status indicator: if power is supplied to a relay, the symbol is displayed
- 6 Status indicator for digital inputs
- 7 Symbol for "device operation locked"

- Display range
  - -99999 to +99999 for measured values
  - 0 to 9999999 for counter values
- Signalization
  - Relay activation
  - Overrange/underrange

#### Operating elements

Jog/shuttle dial

### Remote operation

#### Configuration

The device can be configured with the ReadWin 2000 PC software.

#### Interface

CDI interface at device; connection to PC via USB box (see "Accessories")

RS232 interface at device; connection with serial interface cable (see "Accessories")

---

## Certificates and approvals

|                                       |                                                                                                                                                                                                                                                    |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CE mark</b>                        | The product meets the requirements of the harmonized European standards. As such, it complies with the legal specifications of the EC directives. The manufacturer confirms successful testing of the product by affixing to it the CE-mark.       |
| <b>Ex approval</b>                    | Information about currently available Ex versions (ATEX, FM, CSA. etc.) can be supplied by your Endress+Hauser sales organization on request. All explosion protection data are given in a separate documentation which is available upon request. |
| <b>Other standards and guidelines</b> | The manufacturer confirms compliance with all the relevant external standards and guidelines.                                                                                                                                                      |

## Ordering information

Detailed ordering information is available for your nearest sales organization [www.addresses.endress.com](http://www.addresses.endress.com) or in the Product Configurator under [www.endress.com](http://www.endress.com) :

1. Click Corporate
2. Select the country
3. Click Products
4. Select the product using the filters and search field
5. Open the product page

The Configuration button to the right of the product image opens the Product Configurator.



### Product Configurator - the tool for individual product configuration

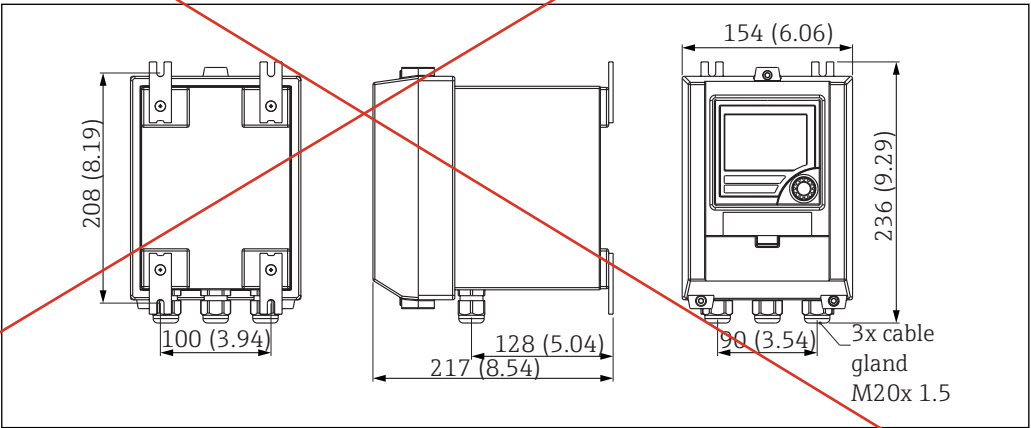
- Up-to-the-minute configuration data
- Depending on the device: Direct input of measuring point-specific information such as measuring range or operating language
- Automatic verification of exclusion criteria
- Automatic creation of the order code and its breakdown in PDF or Excel output format
- Ability to order directly in the Endress+Hauser Online Shop

## Accessories

Various accessories, which can be ordered with the device or subsequently from Endress+Hauser, are available for the device. Detailed information on the order code in question is available from your local Endress+Hauser sales center or on the product page of the Endress+Hauser website: [www.endress.com](http://www.endress.com).

### Device-specific accessories

| Designation                                                                                                | Order No.  |
|------------------------------------------------------------------------------------------------------------|------------|
| PC configuration software ReadWin 2000 and serial configuration cable with 3.5 mm jack plug for RS232 port | RIA452A-VK |
| PC configuration software ReadWin 2000 and serial configuration cable for USB port with CDI connector      | TXU10-AA   |
| Field housing in IP65 →  7,  13                                                                            | 51009957   |
| Current simulator active 4-20mA 1-channel, compact housing, 9V block battery                               | SONDST-S1  |



A0033026

7 Field housing dimensions

## Supplementary documentation

- System components and data manager - solutions to complete your measuring point:  
FA00016K/09
- Brief Operating Instructions for process indicator RIA452: KA00264R/09  
Operating Instructions for process indicator RIA452: BA00265R/09
- Ex-related additional documentation:  
ATEX II(1)GD: XA00053R/09/a3

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[www.addresses.endress.com](http://www.addresses.endress.com)

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